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移動

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- $\beta \longrightarrow K_{3,4}^{\mathrm{int}}$
  $\delta \longrightarrow K_{3,5}^{\mathrm{int}}$
- $\Pi_{3,4} \longrightarrow K_{3,4}^{\pi}$
  $\Pi_{3,5} \longrightarrow K_{3,5}^{\pi}$

NACS: $\hat{E}.\hat{G}$ $(x.y)$ $\tilde{E}.\tilde{G}$ $(x.y)$

一般の座標: $(u.v)$ $E.F.G$

- 1〜3 英語
- 退化次数 n 特異点の セクション を移動 (§4 の初め)

Intrinsic invariants of $(3,4)$-cuspidal edges 12月のRIMSに間に合うように，11月中に投稿する！

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Abstract

$(3,4)$-cuspidal edges naturally appear on parallel surfaces of ellipsoids and other surfaces. For cuspidal edges, the sign of the singular curvature is related to the local shape, whereas for $(3,4)$-cuspidal edges, this sign is not uniquely determined. In this paper, we construct two intrinsic invariants $\kappa_{3,4}^{\mathrm{int}}$ and $\kappa_{3,5}^{\mathrm{int}}$ from the first fundamental form for 3-type edges, which include $(3,4)$-cuspidal edges and $(3,5)$-cuspidal edges. Using these invariants, we give criteria for the boundedness of Gaussian curvature, the convexity and saddle-shapedness of generic $(3,4)$-cuspidal edges, and intrinsic criteria for generic $(3,4)$-cuspidal edges and $(3,5)$-cuspidal edges. Furthermore, we show that these singularity types are preserved under local isometries under suitable assumptions.

1 Introduction

The problem of determining what geometric information can be obtained from the first fundamental form of a surface with singularities is an important one and has been studied by Hasegawa et al. [5], among others. In particular, we investigate to what extent the singularity types, the behavior of Gaussian curvature, and the local shapes near singular points are determined by the first fundamental form. One problem concerning singularity

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types is to determine whether two locally isometric frontals $f, \bar{f}$ are $\mathcal{A}$-equivalent as map-germs at corresponding singular points $p, \bar{p}$. In what follows, we refer to the property that such $\mathcal{A}$-equivalence holds under assumptions appropriate to a given class of singularities as property-$\mathcal{A}$. In particular, for the singularities considered in this paper, we ask whether the following implication holds:

$$\boxed{f, \bar{f} \text{ are locally isometric, and } p, \bar{p} \text{ are generic } \Longrightarrow\ f \sim_{\mathcal{A}} \bar{f}}. \tag{1.1}$$

As a previous result on criteria for singularity types, Kokubu–Rossman–Saji–Umehara–Yamada [1] gave criteria for cuspidal edges and swallowtails at non-degenerate singular points on wave fronts.

Regarding isometric deformations, Naokawa–Umehara–Yamada [2] showed that a generic real analytic cuspidal edge can be isometrically deformed, preserving the limiting normal curvature, into a cuspidal edge whose singular curve lies in a plane. Furthermore, Honda–Naokawa–Saji–Umehara–Yamada [3] extended the isometric duality preserving the first fundamental form and the image of the singular set to generalized cuspidal edges, including cuspidal cross caps and $(2,5)$-cuspidal edges. These studies show that the extrinsic shapes of surfaces may change under deformations preserving the first fundamental form.

On the other hand, concerning the problem of determining singularity types from the first fundamental form, Honda–Naokawa–Umehara–Yamada [8, Proposition 4] gave intrinsic criteria for cuspidal edges, swallowtails, and cuspidal cross caps under the assumption that the limiting normal curvature does not vanish at a non-degenerate singular point. Furthermore, Honda–Saji [9, Corollary 4.5] showed that $(2,5)$-cuspidal edges satisfy property-$\mathcal{A}$ under the assumption that the singular point is of the first kind and the limiting normal curvature does not vanish.

However, Hattori–Honda–Morimoto [10, Corollary 4.9] gave an example showing that property-$\mathcal{A}$ does not necessarily hold even for locally isometric generic frontals. Their counterexample shows that a $(4,5)$-cuspidal edge and a $(4,5,+7)$-cuspidal edge, which are not $\mathcal{A}$-equivalent, can be locally isometric. Table 1 summarizes the previous results and the singularity types considered in this paper.

Table 1: Preservation of singularity types under local isometries and counterexamples under the stated assumptions.

Singularity type	Property-$\mathcal{A}$	Reference
Cuspidal edges, swallowtails, and cuspidal cross caps	Holds	Honda–Naokawa–Umehara–Yamada (2020) [8]
$(2,5)$-cuspidal edges	Holds	Honda–Saji (2019) [9]
$(4,5)$- and $(4,5,\pm 7)$-cuspidal edges	Fails	Hattori–Honda–Morimoto (2025) [10]
$(3,4)$- and $(3,5)$-cuspidal edges	Holds	This paper (Corollary D)

The statements concerning property-$\mathcal{A}$ in the table hold under the assumptions of the respective references. Thus, singularity types are not necessarily preserved under local isometries. To clarify when such preservation holds, it is important to obtain conditions that characterize singularity types in terms of the first fundamental form. In this paper, we investigate this problem for generic $(3,4)$-cuspidal edges and $(3,5)$-cuspidal edges.

These singularities also raise geometric questions concerning their shapes and curvatures. For example, $(3,4)$-cuspidal edges naturally appear on parallel surfaces of ellipsoids (Figure 1). For cuspidal edges, it is known that the singular curvature is an intrinsic invariant and that its sign is related to the local shape of the cuspidal edge [7][12]. However, for $(3,4)$-cuspidal edges, the sign of the singular curvature depends on the orientation of the null direction, and hence its sign alone cannot provide an orientation-independent criterion for convexity and saddle-shapedness. Therefore, we need to introduce an intrinsic invariant that characterizes the local shape of $(3,4)$-cuspidal edges.

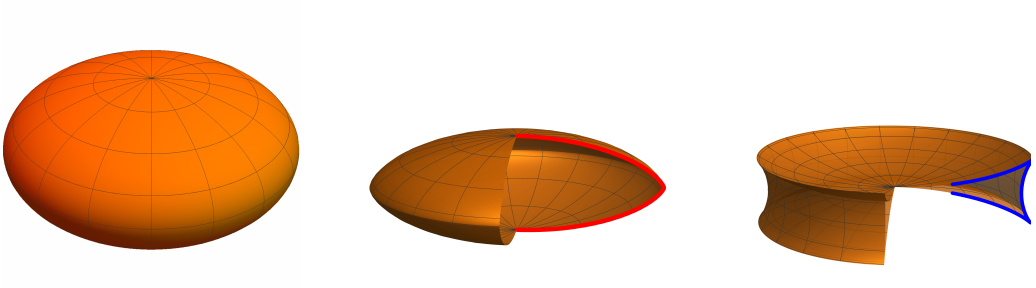


Figure 1: A rotational ellipsoid (left) and two parallel surfaces (center and right). The red line shows a $(3,4)$-cuspidal edge. The blue line shows a cuspidal edge.

Furthermore, Martins–Saji–Umehara–Yamada [6] introduced the product curvature $\kappa_\Pi := \kappa_\nu \kappa_c$, defined as the product of the cuspidal curvature κ_c and the limiting normal curvature κ_ν, and showed that it is an intrinsic invariant related to the boundedness of Gaussian curvature. Regarding classes of singularities, Martins–Saji–Santos–Teramoto [4] introduced m-type edges as a generalization of cuspidal edges and defined the $(m, m+i)$-cuspidal curvatures $\omega_{m,m+i}$ as geometric invariants of m-type edges. In this paper, we focus on 3-type edges, which include $(3,4)$-cuspidal edges and $(3,5)$-cuspidal edges.

Our basic approach is to investigate the behavior of the Gaussian curvature K, which is determined by the first fundamental form, near the singular curve. In the normalized adapted coordinate system (u,v) constructed in Section 3, we assume that the singular set is given by $\{v=0\}$ and consider the expansion of v^2K. We define two quantities $\kappa^{\mathrm{int}}_{3,4}$ and $\kappa^{\mathrm{int}}_{3,5}$ as combinations of higher-order derivatives of the coefficients of the first fundamental form appearing in this expansion. In particular, we have

$$v^2K(u,v) = \frac{1}{3}\kappa^{\mathrm{int}}_{3,4}(u) + O(v), \qquad \text{if } \kappa^{\mathrm{int}}_{3,4} \equiv 0, \quad v^2K(u,v) = \frac{1}{6}\kappa^{\mathrm{int}}_{3,5}(u)v + O(v^2).$$

Here, $\kappa^{\mathrm{int}}_{3,4}$ is independent of the orientation of the null direction, while $\kappa^{\mathrm{int}}_{3,5}$ changes sign when the null direction is reversed. However, $|\kappa^{\mathrm{int}}_{3,5}|$ and the vanishing or non-vanishing of $\kappa^{\mathrm{int}}_{3,5}$ are independent of this orientation. Thus, these quantities provide an intrinsic criterion for the boundedness of Gaussian curvature.

Furthermore, to obtain criteria for singularity types of 3-type edges, we consider the previously introduced $(3,5)$-cuspidal curvature $\omega_{3,5}$. In general, this quantity depends on the choice of the null vector field and is not well-defined without the lower-order vanishing condition $\omega_{3,4} \equiv 0$. In this paper, we introduce a corrected quantity $\widehat{\omega}_{3,5}$ to eliminate this dependence and define the $(3,4)$-product curvature $\kappa^{\Pi}_{3,4}$ and the $(3,5)$-product curvature $\kappa^{\Pi}_{3,5}$ as products with the signed limiting normal curvature. These quantities are related to $\kappa^{\mathrm{int}}_{3,4}$ and $\kappa^{\mathrm{int}}_{3,5}$ by

$$\kappa^{\Pi}_{3,4} = \frac{1}{2}\kappa^{\mathrm{int}}_{3,4}, \qquad \text{if } \omega_{3,4} \equiv 0, \quad \kappa^{\Pi}_{3,5} = \frac{1}{2}\kappa^{\mathrm{int}}_{3,5}.$$

These relations enable us to use $\kappa^{\mathrm{int}}_{3,4}$ and $\kappa^{\mathrm{int}}_{3,5}$ to obtain intrinsic criteria for singularity types. In particular, an advantage of $\kappa^{\mathrm{int}}_{3,5}$ is that it can be defined without imposing the condition $\omega_{3,4} \equiv 0$.

Based on these observations, we use $\kappa^{\mathrm{int}}_{3,4}$ and $\kappa^{\mathrm{int}}_{3,5}$ to investigate (i) the boundedness of Gaussian curvature, (ii) the convexity and saddle-shapedness of $(3,4)$-cuspidal edges, and (iii) intrinsic criteria for generic $(3,4)$-cuspidal edges and $(3,5)$-cuspidal edges. Furthermore, using (iii), we show that these singularity types are preserved under local isometries.

We now state the main results of this paper. Let Σ be a two-dimensional manifold.

Proposition A. *Let $f : \Sigma \to \mathbb{R}^3$ be a frontal with a $(3,4)$-cuspidal edge at $p \in \Sigma$. Then the following statements hold:*

(1) *If $\kappa^{\mathrm{int}}_{3,4}(p) > 0$, then f is convex at p.*

(2) *If $\kappa^{\mathrm{int}}_{3,4}(p) < 0$, then f is saddle-shaped at p.*

(ジェネリックな場合は、
必要十分にできる。(系～参照))

Theorem B. *Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, let K denote the Gaussian curvature of f, and let $S(f)$ denote the singular set of f. Assume that f is a 3-type edge at $p \in \Sigma$. Then, for the quantities $\kappa^{\mathrm{int}}_{3,4}$ and $\kappa^{\mathrm{int}}_{3,5}$ defined in Section 3, the following conditions are equivalent:*

(1) *There exists a neighborhood U of p such that K is bounded on $U \setminus S(f)$.*

(2) *There exists a neighborhood U of p such that $\kappa^{\mathrm{int}}_{3,4} \equiv \kappa^{\mathrm{int}}_{3,5} \equiv 0$ on $S(f) \cap U$.*

Theorem C. *Let $f : \Sigma \to \mathbb{R}^3$ be a frontal with a 3-type edge at $p \in \Sigma$. Assume that p is generic. Then the following conditions are equivalent:*

(1) *f has a $(3,4)$-cuspidal edge at p.*

(2) *$\kappa^{\mathrm{int}}_{3,4}(p) \neq 0$.*

Moreover, the following conditions are also equivalent:

(1) *f has a $(3,5)$-cuspidal edge at p.*

(2) *There exists a neighborhood U of p such that $\kappa^{\mathrm{int}}_{3,4} \equiv 0$ on $S(f) \cap U$ and $\kappa^{\mathrm{int}}_{3,5}(p) \neq 0$.*

Corollary D. *Let $f : \Sigma \to \mathbb{R}^3$ and $g : \overline{\Sigma} \to \mathbb{R}^3$ be isometric frontals, and let $p \in \Sigma$ and $\overline{p} \in \overline{\Sigma}$ be corresponding generic singular points. Then, for $j = 4, 5$, the following conditions are equivalent:*

(1) *f has a $(3,j)$-cuspidal edge at p.*

(2) *g has a $(3,j)$-cuspidal edge at $\overline{p}$.*

In Section 2, we recall the notions from previous studies. In Section 3, we construct the normalized adapted coordinate system (NACS) used in this paper and introduce intrinsic invariants using this coordinate system. We then prove Proposition A (Proposition 3.12) and Theorem B (Theorem 3.15) using these invariants. In Section 4, we introduce singular points of degeneracy order n and prove criteria for $(3,4)$-cuspidal edges and $(3,5)$-cuspidal edges. Using these criteria, we derive Theorem C (Theorem 5.12) and Corollary D (Corollary 5.19) in Section 5.

2 Preliminaries

Throughout this paper, let Σ and $\overline{\Sigma}$ be two-dimensional manifolds, and let $\mathbb{S}^2 \subset \mathbb{R}^3$ denote the unit sphere centered at the origin. We denote by $\langle\ ,\ \rangle$ and $\|\ \|$ the standard Euclidean inner product and norm on $\mathbb{R}^3$, respectively. Unless otherwise stated, all maps, functions, and vector fields are assumed to be of class C^∞. Partial derivatives with respect to local coordinates (u, v) are denoted by subscripts. For example, for $j \in \mathbb{N}$, we write $f_{v^j} := \partial^j f / \partial v^j$.

2.1 Definitions of 3-type edges and invariants

A point $p \in \Sigma$ is called a *singular point* of a map $f : \Sigma \to \mathbb{R}^3$ if f is not an immersion at p, that is, if $\mathrm{rank}(df)_p < 2$. The set of all singular points of f is denoted by

$$S(f) := \{q \in \Sigma \mid \mathrm{rank}(df)_q < 2\}.$$

To define singularity types in terms of $\mathcal{A}$-equivalence, we first recall the following equivalence relation.

Definition 2.1. Two map-germs $f : (\Sigma, p) \to (\mathbb{R}^3, f(p))$ and $\overline{f} : (\overline{\Sigma}, \overline{p}) \to (\mathbb{R}^3, \overline{f}(\overline{p}))$ are said to be *$\mathcal{A}$-equivalent* if there exist two diffeomorphism-germs $\varphi : (\Sigma, p) \to (\overline{\Sigma}, \overline{p})$ and $\Phi : (\mathbb{R}^3, f(p)) \to (\mathbb{R}^3, \overline{f}(\overline{p}))$ such that

$$\Phi \circ f \circ \varphi^{-1} = \overline{f}.$$

The $(3,4)$-cuspidal edge and the $(3,5)$-cuspidal edge are defined as follows.

Definition 2.2. Let $f : \Sigma \to \mathbb{R}^3$ be a map and $p \in \Sigma$. A point p is called a $(3,4)$-*cuspidal edge* of f if the map-germ $f : (\Sigma, p) \to (\mathbb{R}^3, f(p))$ is $\mathcal{A}$-equivalent to the map-germ at the origin of

$$f^{\mathrm{std}}_{3,4} : \mathbb{R}^2 \ni (u, v) \longmapsto \left(v^3, v^4, u\right) \in \mathbb{R}^3.$$

Similarly, p is called a $(3,5)$-*cuspidal edge* of f if the map-germ of f at p is $\mathcal{A}$-equivalent to the map-germ at the origin of

$$f^{\mathrm{std}}_{3,5} : \mathbb{R}^2 \ni (u, v) \longmapsto \left(v^3, v^5, u\right) \in \mathbb{R}^3.$$

More generally, the map-germ $f : (\Sigma, p) \to (\mathbb{R}^3, f(p))$ is called an *m-type edge* if it is $\mathcal{A}$-equivalent to the map-germ at the origin of

$$\mathbb{R}^2 \ni (u, v) \longmapsto \left(v^m, v^{m+1}a(u, v), u\right) \in \mathbb{R}^3$$

for some smooth function $a(u, v)$[4].

By definition, both $(3,4)$-cuspidal edges and $(3,5)$-cuspidal edges are 3-type edges. We call $f^{\mathrm{std}}_{3,4}$ and $f^{\mathrm{std}}_{3,5}$ the **standard $(3,4)$-cuspidal edge** and the **standard $(3,5)$-cuspidal edge**, respectively (see Figure 2). The class of 2-type edge germs coincides with that of generalized cuspidal edge germs introduced in [3].

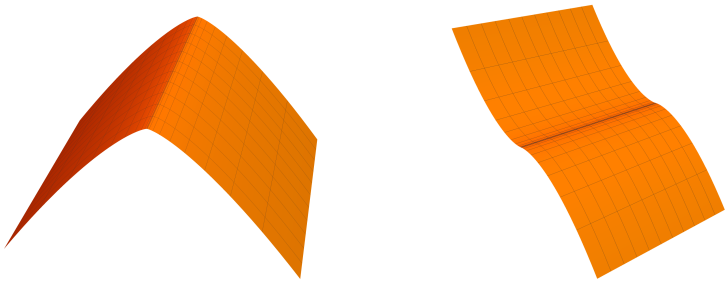


Figure 2: The standard $(3,4)$-cuspidal edge (left) and the standard $(3,5)$-cuspidal edge (right)

We define the invariants and intrinsic invariants used in this paper as follows.

Definition 2.3 ([10]). Let $f : \Sigma \to \mathbb{R}^3$ be a C^∞-map and $p \in \Sigma$. Let $J(f, p)$ denote a value determined by f and p.

(1) The quantity $J(f, p)$ is said to be a *geometric invariant* if

$$|J(f, p)| = \left|J\left(T \circ f \circ \varphi^{-1}, \varphi(p)\right)\right|$$

holds for any isometry T of $\mathbb{R}^3$ and any diffeomorphism φ of Σ.

(2) Let ds^2_f and $ds^2_{\overline{f}}$ be the first fundamental forms of $f : \Sigma \to \mathbb{R}^3$ and $\overline{f} : \overline{\Sigma} \to \mathbb{R}^3$, respectively. We say that f and $\overline{f}$ are *isometric* near p and $\overline{p}$ if there exists a diffeomorphism-germ $\varphi : (\Sigma, p) \to (\overline{\Sigma}, \overline{p})$ such that

$$\varphi^* ds^2_{\overline{f}} = ds^2_f.$$

(3) A geometric invariant $J(f, p)$ is said to be an *intrinsic invariant* if

$$|J(f, p)| = |J(\overline{f}, \overline{p})|$$

holds for any map $\overline{f} : \overline{\Sigma} \to \mathbb{R}^3$ that is isometric to f near p and $\overline{p}$, and for which $J(\overline{f}, \overline{p})$ is defined. A geometric invariant is said to be an *extrinsic invariant* if it is not an intrinsic invariant.

For regular surfaces, the Gaussian curvature is an intrinsic invariant, since it is determined solely by the first fundamental form. On the other hand, the mean curvature is an extrinsic invariant, since its value may change under isometric deformations that preserve the first fundamental form.

2.2 Criteria for 3-type edges

To state a criterion for 3-type edges, we first recall the notions of singular directions, null directions, singular points of the first kind, and frontals.

Let $f : \Sigma \to \mathbb{R}^3$ be a C^∞-map, and let $p \in S(f)$ be a singular point. Assume that the singular set $S(f)$ is a 1-dimensional submanifold (a regular curve) of Σ near p.

Let $q \in S(f)$. A non-zero vector $\xi_q \in T_qS(f)$ is called a *singular vector* at q, and the one-dimensional subspace $T_qS(f) = \mathrm{span}_{\mathbb{R}}\{\xi_q\} \subset T_q\Sigma$ is called the *singular direction* at q. A smooth non-vanishing vector field ξ defined on a neighborhood $U \subset \Sigma$ of $S(f)$ is called a *singular vector field* if $\xi_q \in T_qS(f)$ holds for every $q \in S(f) \cap U$.

Furthermore, suppose that $q \in S(f)$ is a *rank one singular point*, that is, $\mathrm{rank}(df)_q = 1$. A non-zero vector $\eta_q \in \ker(df)_q$ is called a *null vector* at q, and the one-dimensional subspace $\ker(df)_q = \mathrm{span}_{\mathbb{R}}\{\eta_q\} \subset T_q\Sigma$ is called the *null direction* at q. A smooth non-vanishing vector field η defined on a neighborhood $U \subset \Sigma$ of $S(f)$ is called a *null vector field* if $\eta_q \in \ker(df)_q$ holds for every $q \in S(f) \cap U$.

Definition 2.4. Let $f : \Sigma \to \mathbb{R}^3$ be a C^∞-map, and assume that the singular set $S(f)$ is a regular curve. Let $p \in S(f)$ be a rank one singular point. If the singular direction and the null direction at p are distinct, that is, if $T_pS(f) \neq \ker(df)_p$, then p is called a *singular point of the first kind*. Otherwise, p is called a *singular point of the second kind*.

Definition 2.5. A map $f : \Sigma \to \mathbb{R}^3$ is called a *frontal* if, for each point $p \in \Sigma$, there exist an open neighborhood U of p and a smooth map $\nu : U \to S^2$ such that

$$\langle df_q(X_q), \nu(q) \rangle = 0$$

holds for every point $q \in U$ and every tangent vector $X_q \in T_q\Sigma$. Such a ν is called a *unit normal vector field* along f. Moreover, f is called a *front* or a *wave front* if, for each point $p \in \Sigma$, there exists such a ν for which

$$L := (f, \nu) : U \to \mathbb{R}^3 \times \mathbb{S}^2$$

is an immersion. We say that f is a *front at* p if there exists such a ν for which L is an immersion at p.

The following criterion for 3-type edges is then obtained.

Fact 2.6 ([4, Proposition 2.2])**.** *Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, and let $p \in \Sigma$ be a rank one singular point. Assume that the singular set $S(f)$ is a regular curve near p, and let ξ be a singular vector field. Assume also that p is a singular point of the first kind. Then, f is a 3-type edge at p if and only if there exists a null vector field η satisfying*

$$\eta f = \eta^2 f = 0 \quad \text{on } S(f), \tag{2.1}$$

$$\xi f(p) \quad \text{and} \quad \eta^3 f(p) \quad \text{are linearly independent.} \tag{2.2}$$

Here, for a vector field ζ and a map f, the symbol $\zeta^i f$ stands for the i-times directional derivative of f by ζ.

In what follows, we fix a singular vector field ξ and a null vector field η satisfying the conditions in Fact 2.6.

3 Intrinsic invariants of 3-type edges and their applications

In this section, we first construct a normalized adapted coordinate system for 3-type edges and use it to express the singular curvature in terms of the coefficients of the first fundamental form. Next, we introduce $\kappa^{\mathrm{int}}_{3,4}, \kappa^{\mathrm{int}}_{3,5}$ using higher-order derivatives of these coefficients and show that they are intrinsic invariants. Furthermore, using these invariants, we investigate the local shape of $(3,4)$-cuspidal edges and the boundedness of Gaussian curvature, and prove Proposition 3.12 and Theorem 3.15.

3.1 Normalized adapted coordinate system

In this subsection, we construct a normalized adapted coordinate system to study higher-order derivatives of the first fundamental form near 3-type edges, and establish its existence and transformation formulas between such coordinate systems.

Definition 3.1. Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, and let $p \in \Sigma$. Let $(U; x, y)$ be a local coordinate system centered at p such that

$$S(f) \cap U = \{y = 0\}.$$

We call (x, y) a normalized adapted coordinate system (NACS for short) if it satisfies the following conditions:

$$\begin{aligned} f_y(x,0) = f_{yy}(x,0) &= \mathbf{0}, \\ \|f_x(x,0)\| = \|f_{y^3}(x,0)\| &= 1, \\ \langle f_x, f_y\rangle(x,y) &\equiv 0. \end{aligned}$$

By the definition of a NACS, we have $\langle f_x, f_{y^3}\rangle(x,0) = \langle f_x, f_y\rangle_{yy}(x,0) = 0$. Moreover, if we denote the coefficients of the first fundamental form in a NACS (x, y) by

$$\hat{E} = \langle f_x, f_x\rangle, \qquad \hat{F} = \langle f_x, f_y\rangle, \qquad \hat{G} = \langle f_y, f_y\rangle$$

then the definition of a NACS is equivalent to the following conditions:

$$\hat{E}(x,0) = 1, \qquad \hat{F}(x,y) \equiv 0, \tag{3.1}$$

$$\hat{G}(x,0) = \hat{G}_y(x,0) = \hat{G}_{yy}(x,0) = \hat{G}_{y^3}(x,0) = 0, \tag{3.2}$$

$$\hat{G}_{y^4}(x,0) = 6. \tag{3.3}$$

In fact, under the conditions $f_y(x,0) = f_{yy}(x,0) = \mathbf{0}$, we have $\hat{G}_{y^4}(x,0) = 6\|f_{y^3}(x,0)\|^2$. Conversely, $\hat{G}(x,0) = \hat{G}_{yy}(x,0) = 0$ implies $f_y(x,0) = f_{yy}(x,0) = \mathbf{0}$, and the remaining conditions can be verified immediately.

Proposition 3.2. *Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, and let $p \in \Sigma$. Then the following conditions are equivalent.*

(1) *f is a 3-type edge at p.*

(2) *There exists a NACS in a neighborhood of p.*

To prove this proposition, we prepare the following lemma 3.3.

Lemma 3.3. *Let $f : \Sigma \to \mathbb{R}^3$ be a frontal with a 3-type edge at $p \in \Sigma$, and let ν be a unit normal vector field along f. Then the following hold:*

- *There exists a local coordinate system $(U; u, v)$ centered at p satisfying the following conditions near p:*
  - *(1) $S(f) \cap U = \{(u, v) \in U \mid v = 0\}$.*
  - *(2) ∂_v gives the null direction along $S(f)$, and $f_v(u,0) = f_{vv}(u,0) = \mathbf{0}$.*
  - *(3) For each $(u,0) \in S(f) \cap U$, the vectors $\{f_u(u,0), f_{v^3}(u,0), \nu(u,0)\}$ form an orthonormal basis of $\mathbb{R}^3$.*
- *In particular, the above coordinate conditions are equivalent to the following conditions along the singular set:*

$$\begin{aligned} E(u,0) &= 1, \\ F(u,0) = F_v(u,0) &= F_{vv}(u,0) = 0, \\ G(u,0) = G_v(u,0) &= G_{vv}(u,0) = G_{v^3}(u,0) = 0, \\ G_{v^4}(u,0) &= 6 \end{aligned}$$

where $E = \langle f_u, f_u\rangle$, $F = \langle f_u, f_v\rangle$, and $G = \langle f_v, f_v\rangle$.

Proof. First, we prove the existence of the coordinate system. Since f is a 3-type edge at p, by Fact 2.6, we can take local coordinates (s,t) near p such that

$$S(f) = \{t = 0\}, \qquad f_t(s,0) = f_{tt}(s,0) = \mathbf{0}$$

and

$$\operatorname{rank}\bigl(f_s, f_{t^3}\bigr)(p) = 2$$

hold. Since the image of the singular curve $s \mapsto f(s,0)$ is regular, we may reparametrize s by arc length so that

$$\|f_s(s,0)\| = 1 \tag{3.4}$$

holds. Moreover, by shrinking the neighborhood if necessary, $f_s(s,0)$ and $f_{t^3}(s,0)$ are linearly independent along the singular curve. By division lemma, there exists a smooth map φ such that

$$f_t(s,t) = t^2\varphi(s,t) \tag{3.5}$$

holds. In particular, $f_{t^3}(s,0) = 2\varphi(s,0)$.

Using a smooth function $a(s)$ and a positive smooth function $r(s)$, we define new coordinates (u,v) by

$$u = s + a(s)t^3, \qquad v = r(s)t. \tag{3.6}$$

On $t = 0$, the Jacobian matrix is given by

$$\frac{\partial(u,v)}{\partial(s,t)}(s,0) = \begin{pmatrix} 1 & 0 \\ 0 & r(s) \end{pmatrix}$$

Since $r(s) > 0$, the coordinate transformation (3.6) is a local diffeomorphism after shrinking the neighborhood sufficiently. Moreover, since $v = 0$ if and only if $t = 0$, the singular set in the new coordinates is also given by

$$S(f) = \{v = 0\}. \tag{3.7}$$

By the chain rule and (3.5), we can write $f_v(u,v) = v^2\widetilde{\varphi}(u,v)$ in the new coordinates. Thus,

$$f_v(u,0) = f_{vv}(u,0) = \mathbf{0} \tag{3.8}$$

and ∂_v gives the null direction along the singular curve. Furthermore, applying the chain rule along the singular curve, we obtain

$$f_u(u,0) = f_s(s,0), \tag{3.9}$$

$$f_{v^3}(u,0) = \frac{1}{r(s)^3}\left\{f_{t^3}(s,0) - 6a(s)f_s(s,0)\right\}. \tag{3.10}$$

Here, we set

$$a(s) := \frac{1}{6}\left\langle f_s(s,0), f_{t^3}(s,0)\right\rangle, \tag{3.11}$$

$$r(s) := \left\|f_{t^3}(s,0) - \left\langle f_s(s,0), f_{t^3}(s,0)\right\rangle f_s(s,0)\right\|^{1/3}. \tag{3.12}$$

Since $f_s(s,0)$ and $f_{t^3}(s,0)$ are linearly independent and (3.4) holds, r is a positive smooth function. By (3.9)–(3.12), we obtain

$$\|f_u(u,0)\| = 1, \qquad \langle f_u, f_{v^3}\rangle(u,0) = 0, \qquad \|f_{v^3}(u,0)\| = 1.$$

Moreover, it follows from (3.5) and $\langle f_t, \nu\rangle \equiv 0$ that $f_{t^3}(s,0)$ is orthogonal to $\nu(s,0)$. Since $f_s(s,0)$ is also orthogonal to $\nu(s,0)$, it follows from (3.10) that $f_{v^3}(u,0)$ is orthogonal to $\nu(u,0)$. Therefore,

$$\{f_u(u,0), f_{v^3}(u,0), \nu(u,0)\}$$

forms an orthonormal basis of $\mathbb{R}^3$. Combining this with (3.7) and (3.8), we obtain the existence of the desired coordinate system.

Next, we prove the forward implication. Suppose that $f_v = f_{vv} = \mathbf{0}$ holds along the singular curve and that $\{f_u, f_{v^3}, \nu\}$ forms an orthonormal basis. Then,

$$E(u,0) = \langle f_u, f_u\rangle(u,0) = 1$$

holds.

For $F = \langle f_u, f_v \rangle$, the Leibniz rule yields

$$F_{v^j} = \sum_{k=0}^{j} \binom{j}{k} \langle f_{uv^k}, f_{v^{j-k+1}} \rangle .$$

Combining this with $f_v(u,0) = f_{vv}(u,0) = \mathbf{0}$, we obtain

$$F(u,0) = F_v(u,0) = 0, \qquad F_{vv}(u,0) = \langle f_u, f_{v^3} \rangle (u,0) = 0.$$

Similarly, for $G = \langle f_v, f_v \rangle$, we have

$$G_{v^j} = \sum_{k=0}^{j} \binom{j}{k} \langle f_{v^{k+1}}, f_{v^{j-k+1}} \rangle .$$

Therefore, $G = G_v = G_{vv} = G_{v^3} = 0$ along the singular curve, and

$$G_{v^4}(u,0) = 6\|f_{v^3}(u,0)\|^2 = 6$$

holds.

Finally, we prove the converse implication. Assume that the coefficients of the first fundamental form satisfy the conditions in the lemma. Since $G(u,0) = 0$, we have $f_v(u,0) = \mathbf{0}$, and

$$G_{vv}(u,0) = 2\|f_{vv}(u,0)\|^2 = 0$$

implies $f_{vv}(u,0) = \mathbf{0}$. Moreover, by division lemma, there exist smooth real-valued functions A, B such that

$$F = v^3 A, \qquad G = v^4 B, \qquad B(u,0) = \frac{1}{4}.$$

Since $E(u,0) = 1$ and $B(u,0) = 1/4$, we have

$$EG - F^2 = v^4(EB - v^2 A^2) > 0 \qquad (v \neq 0)$$

in a sufficiently small neighborhood. Thus, $S(f) \cap U = \{v = 0\}$, and ∂_v gives the null direction along the singular set. Furthermore, on $v = 0$,

$$\|f_u\|^2 = E = 1, \qquad \langle f_u, f_{v^3} \rangle = F_{vv} = 0, \qquad 6\|f_{v^3}\|^2 = G_{v^4} = 6$$

hold. Since f is a frontal, we have $\langle f_v, \nu \rangle \equiv 0$. Differentiating this identity twice with respect to v, we obtain $\langle f_{v^3}, \nu \rangle (u,0) = 0$. Moreover, $\langle f_u, \nu \rangle = 0$, and hence $\{f_u, f_{v^3}, \nu\}$ forms an orthonormal basis of $\mathbb{R}^3$. □

Using the above lemma, we prove the existence of a NACS.

Proof of Proposition 3.2. First, we prove (1) $\Longrightarrow$ (2). Since f is a 3-type edge at p, we take a coordinate system (u,v) satisfying the conditions in Lemma 3.3, and denote the coefficients of its first fundamental form by E, F, G. Then, by the division lemma, there exist smooth real-valued functions A, B such that

$$F = v^3 A, \qquad G = v^4 B, \qquad B(u,0) = \frac{1}{4}. \tag{3.13}$$

Let $(x(u,v), y(u,v))$ be new coordinates, and set $y = v$. We define x as a local solution of the first-order partial differential equation

$$Fx_u - Ex_v = 0, \qquad x(u,0) = u. \tag{3.14}$$

This is the initial value problem for the partial differential equation

$$x_v - \frac{F}{E} x_u = 0.$$

Indeed, since $F = 0$ and $E = 1$ on $v = 0$, the characteristic direction is transverse to the initial curve $v = 0$. Thus, a local solution exists by the method of characteristics. Since the initial condition implies $x_u(u, 0) = 1$, the coordinate transformation $(u, v) \mapsto (x, y)$ is a local diffeomorphism. We now show that (x, y) satisfies the definition of a NACS.

Writing the inverse transformation as $u = u(x, y)$, $v = y$, we have

$$\partial_x = \frac{1}{x_u}\partial_u, \qquad \partial_y = -\frac{x_v}{x_u}\partial_u + \partial_v.$$

Let $\hat{E}, \hat{F}, \hat{G}$ denote the coefficients of the first fundamental form in the new coordinates. Then, by (3.14), we have

$$\hat{F} = \langle f_x,\, f_y \rangle = \frac{-x_v E + x_u F}{x_u^2} = 0.$$

Moreover, since $x_u(u, 0) = E(u, 0) = 1$, we obtain $\hat{E}(x, 0) = 1$. Furthermore,

$$\hat{G} = \langle f_y,\, f_y \rangle = G - \frac{F^2}{E} = v^4 B - v^6 \frac{A^2}{E}. \tag{3.15}$$

Using $v = y$ and $B(u, 0) = 1/4$ from (3.13), we obtain (3.2) and (3.3). Thus, (x, y) satisfies (3.1)–(3.3), which are equivalent to the conditions in Definition 3.1.

Next, we prove (2) $\implies$ (1). Let (x, y) be a NACS. By the definition of a NACS, we have $S(f) = \{y = 0\}$, and ∂_x and ∂_y give the singular and null directions, respectively. Thus, p is a rank-one singular point of the first kind. Moreover,

$$\|f_x(x, 0)\| = \|f_{y^3}(x, 0)\| = 1 \neq 0, \qquad \langle f_x, f_{y^3} \rangle(x, 0) = \hat{F}_{yy}(x, 0) = 0.$$

Hence, $\operatorname{rank}(f_x, f_{y^3})(p) = 2$. Furthermore, $f_y(x, 0) = f_{y^2}(x, 0) = \mathbf{0}$ holds along the singular curve. Taking $\xi = \partial_x$ and $\eta = \partial_y$, we conclude from Fact 2.6 that f is a 3-type edge at p. □

Remark 3.4. In [5, Proposition 2.25], the existence of a normalized strongly adapted coordinate system (u, v) at an A_2-point of a Kossowski metric is established, satisfying

$$E(u, 0) = 1, \quad E_v(u, 0) = 0, \quad F \equiv 0, \quad G(u, 0) = G_v(u, 0) = 0, \quad G_{vv}(u, 0) = 2.$$

A NACS is the corresponding coordinate system for 3-type edges, with the normalization $G_{v^4}(u, 0) = 6$.

In a NACS, the Taylor expansions of the coefficients of the first fundamental form are given as follows.

Corollary 3.5. *Let $f : \Sigma \to \mathbb{R}^3$ be a frontal with a 3-type edge at $p \in \Sigma$. Let (x, y) be a NACS near p. Then,*

$$\hat{E}_y(x, 0) = \hat{E}_{yy}(x, 0) = 0. \tag{3.16}$$

Moreover, we have

$$\hat{E}(x, y) = 1 + \frac{\hat{E}_{y^3}(x, 0)}{6} y^3 + \frac{\hat{E}_{y^4}(x, 0)}{24} y^4 + \frac{\hat{E}_{y^5}(x, 0)}{120} y^5 + O(y^6), \tag{3.17}$$

$$\hat{G}(x, y) = \frac{1}{4} y^4 + \frac{\hat{G}_{y^5}(x, 0)}{120} y^5 + \frac{\hat{G}_{y^6}(x, 0)}{720} y^6 + O(y^7). \tag{3.18}$$

Proof. Since f is a 3-type edge at p, the division lemma yields $f_y = y^2 \varphi$. Hence,

$$\hat{E}_y = 2\langle f_x, f_{xy} \rangle = O(y^2), \qquad \hat{E}_{yy} = O(y).$$

Thus, $\hat{E}_y = \hat{E}_{yy} = 0$ along $y = 0$. The remaining assertions follow from the equivalent conditions (3.1)–(3.3) for a NACS and the Taylor expansions. □

Finally, we examine how the higher-order derivatives of the coefficients of the first fundamental form transform between two NACSs. These transformation formulas will be used to prove the intrinsicity of $\kappa_{3,4}^{\mathrm{int}}, \kappa_{3,5}^{\mathrm{int}}$ introduced in Section 3.3 (Proposition 3.10).

Let (x, y) and $(\tilde{x}, \tilde{y})$ be two NACSs near p, and write the coordinate transformation as

$$x = x(\tilde{x}, \tilde{y}), \qquad y = y(\tilde{x}, \tilde{y}).$$

Let $\hat{E}, \hat{G}$ and $\tilde{E}, \tilde{G}$ denote the coefficients of the first fundamental form with respect to (x, y) and $(\tilde{x}, \tilde{y})$, respectively. Along the singular set, we have

$$y(\tilde{x}, 0) = y_{\tilde{x}}(\tilde{x}, 0) = 0.$$

Moreover, by the definition of a NACS and the chain rule, we obtain

$$x_{\tilde{y}}(\tilde{x}, 0) = 0, \qquad x_{\tilde{x}}(\tilde{x}, 0) = \pm 1, \qquad y_{\tilde{y}}(\tilde{x}, 0) = \pm 1.$$

Lemma 3.6. *Under the above setting, the following formulas hold along the singular set:*

$$\tilde{E}_{\tilde{y}^3} = y_{\tilde{y}} \hat{E}_{y^3}, \tag{3.19}$$

$$\tilde{E}_{\tilde{y}^4} = \hat{E}_{y^4} + 6 y_{\tilde{y}^2} \hat{E}_{y^3}, \tag{3.20}$$

$$\tilde{E}_{\tilde{y}^5} = y_{\tilde{y}} \hat{E}_{y^5} + 10 y_{\tilde{y}^2} y_{\tilde{y}} \hat{E}_{y^4} + (10 y_{\tilde{y}^3} + 15\{y_{\tilde{y}^2}\}^2 y_{\tilde{y}}) \hat{E}_{y^3}, \tag{3.21}$$

$$\tilde{G}_{\tilde{y}^5} = y_{\tilde{y}} \big(\hat{G}_{y^5} + 120 y_{\tilde{y}^2} \big), \tag{3.22}$$

$$\tilde{G}_{\tilde{y}^6} = \hat{G}_{y^6} + 27 y_{\tilde{y}^2} \hat{G}_{y^5} + 300 y_{\tilde{y}} y_{\tilde{y}^3} + 1170\{y_{\tilde{y}^2}\}^2. \tag{3.23}$$

Here, $\tilde{E}$, $\tilde{G}$, $y_{\tilde{y}}$ are evaluated at $(\tilde{x}, 0)$, while $\hat{E}$, $\hat{G}$ are evaluated at $(x(\tilde{x}, 0), 0)$.

Proof. Comparing the coefficients of the first fundamental forms, we obtain

$$\tilde{E} = \hat{E} x_{\tilde{x}}^2 + \hat{G} y_{\tilde{x}}^2, \qquad 0 = \hat{E} x_{\tilde{x}} x_{\tilde{y}} + \hat{G} y_{\tilde{x}} y_{\tilde{y}}, \qquad \tilde{G} = \hat{E} x_{\tilde{y}}^2 + \hat{G} y_{\tilde{y}}^2.$$

In what follows, when differentiating $\hat{E}, \hat{G}$ with respect to $(\tilde{x}, \tilde{y})$, we regard them as compositions with the coordinate transformation. Applying the Leibniz rule to these identities and differentiating with respect to $\tilde{y}$ up to the required orders, we obtain the following formulas, where $\tilde{E}$, $\tilde{G}$, $y_{\tilde{y}}$ are evaluated at $(\tilde{x}, 0)$ and $\hat{E}$, $\hat{G}$ at $(x(\tilde{x}, 0), 0)$:

$$\begin{aligned}
\tilde{E}_{\tilde{y}^3} &= y_{\tilde{y}} \hat{E}_{y^3} \\
\tilde{E}_{\tilde{y}^4} &= \hat{E}_{\tilde{y}^4} + \hat{G}_{\tilde{y}^4} \, (y_{\tilde{x}})^2 \\
\tilde{E}_{\tilde{y}^5} &= \hat{E}_{\tilde{y}^5} + 2 x_{\tilde{x}} x_{\tilde{x}\tilde{y}^5} + \hat{G}_{\tilde{y}^5} \left(y_{\tilde{x}}\right)^2 + 10 \hat{G}_{\tilde{y}^4} \, y_{\tilde{x}} y_{\tilde{x}\tilde{y}} \\
\tilde{G}_{\tilde{y}^5} &= \big(y_{\tilde{y}}\big)^2 \hat{G}_{\tilde{y}^5} + 10 y_{\tilde{y}} y_{\tilde{y}^2} \, \hat{G}_{\tilde{y}^4} \\
\tilde{G}_{\tilde{y}^6} &= \hat{G}_{\tilde{y}^6} + 12 y_{\tilde{y}} y_{\tilde{y}^2} \hat{G}_{\tilde{y}^5} + 30\big((y_{\tilde{y}^2})^2 + y_{\tilde{y}} y_{\tilde{y}^3}\big) \hat{G}_{\tilde{y}^4}.
\end{aligned}$$

Since both coordinate systems are NACSs, we substitute the conditions (3.1)–(3.3) and (3.16) of Corollary 3.5 along the singular set. Furthermore, by explicitly computing $\hat{E}_{\tilde{y}^4}, \hat{E}_{\tilde{y}^5}, \hat{G}_{\tilde{y}^4}, \hat{G}_{\tilde{y}^5}, \hat{G}_{\tilde{y}^6}$ using the chain rule and substituting the resulting expressions, we obtain (3.19)–(3.23). □

3.2 Intrinsic expression for singular curvature

In this subsection, we define the singular curvature of 3-type edges and express it in terms of the first fundamental form using a NACS.

Definition 3.7. Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, and let $\gamma(t)$ be a parametrization of a singular curve consisting of singular points of the first kind at which f is a 3-type edge. Set $\widehat{\gamma} := f \circ \gamma$ and $\widehat{\nu} := \nu \circ \gamma$. For 3-type edges, we define the singular curvature by

$$\kappa_s(t) := \frac{\det\big(\widehat{\gamma}'(t), \widehat{\gamma}''(t), \widehat{\nu}(t)\big)}{\|\widehat{\gamma}'(t)\|^3}. \tag{3.24}$$

The sign of this singular curvature depends on the orientation of the parametrization of the singular curve and the choice of the unit normal vector field. In what follows, we take a NACS (x, y) and parametrize the singular curve by $\gamma(x) = (x, 0)$. We also choose the orientation of the unit normal vector field so that

$$\nu(x, 0) = f_x(x, 0) \times f_{y^3}(x, 0).$$

Proposition 3.8. *Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, and let (x, y) be a NACS. Then the singular curvature κ_s along the singular curve $y = 0$ is expressed as*

$$\kappa_s(x) = -\frac{1}{2}\hat{E}_{y^3}(x, 0). \tag{3.25}$$

Thus, κ_s is an intrinsic invariant.

Proof. Parametrizing the singular curve by $\gamma(x) = (x, 0)$, we have $\widehat{\gamma}'(x) = f_x(x, 0)$ and $\widehat{\gamma}''(x) = f_{xx}(x, 0)$. By the conditions of a NACS, we have

$$\|f_x(x, 0)\| = \|f_{y^3}(x, 0)\| = 1, \qquad \langle f_x, f_{y^3}\rangle(x, 0) = 0.$$

Moreover, the orientation of the unit normal vector field is chosen so that $\nu(x, 0) = f_x(x, 0) \times f_{y^3}(x, 0)$. Therefore, by the definition of singular curvature (3.24), we obtain

$$\begin{aligned}\kappa_s(x) &= \det(f_x, f_{xx}, \nu)(x, 0) = \left\langle f_x \times f_{xx}, f_x \times f_{y^3}\right\rangle(x, 0)\\ &= \langle f_{xx}, f_{y^3}\rangle(x, 0).\end{aligned} \tag{3.26}$$

On the other hand, since f is a 3-type edge, we have $f_y = f_{yy} = \mathbf{0}$ along the singular curve. Differentiating these identities with respect to x, we obtain $f_{xy} = f_{xyy} = \mathbf{0}$. Thus, differentiating $\hat{E} = \langle f_x, f_x\rangle$ three times with respect to y, we obtain

$$\hat{E}_{y^3}(x, 0) = 2\langle f_x, f_{xy^3}\rangle(x, 0). \tag{3.27}$$

Furthermore, using $\hat{F} = \langle f_x, f_y\rangle \equiv 0$ and computing $\hat{F}_{xyy}$, we obtain

$$0 = \hat{F}_{xyy}(x, 0) = \langle f_{xx}, f_{y^3}\rangle(x, 0) + \langle f_x, f_{xy^3}\rangle(x, 0).$$

Hence, (3.27) yields $\hat{E}_{y^3}(x, 0) = -2\langle f_{xx}, f_{y^3}\rangle(x, 0)$. Substituting this into (3.26), we obtain (3.25). □

3.3 Introduction of intrinsic invariants $\kappa^{\mathrm{int}}_{3,4}$ and $\kappa^{\mathrm{int}}_{3,5}$

In this subsection, we define $\kappa^{\mathrm{int}}_{3,4}, \kappa^{\mathrm{int}}_{3,5}$ using higher-order derivatives of the coefficients of the first fundamental form in a NACS and prove their intrinsicity. These invariants are related to the expansion of the Gaussian curvature K near the singular curve. In fact, in subsection 3.5, we show that the constant term of y^2K is expressed in terms of $\kappa^{\mathrm{int}}_{3,4}$, while the linear term is expressed in terms of $\kappa^{\mathrm{int}}_{3,4}$ and $\kappa^{\mathrm{int}}_{3,5}$ (Lemma 3.16).

Definition 3.9. Let $f : \Sigma \to \mathbb{R}^3$ be a frontal with a 3-type edge at $p \in \Sigma$. Let (x, y) be a NACS centered at $p = (0, 0)$. We define the quantities $\kappa^{\mathrm{int}}_{3,4}(p)$ and $\kappa^{\mathrm{int}}_{3,5}(p)$ by

$$\kappa^{\mathrm{int}}_{3,4}(p) := -\hat{E}_{y^4}(0, 0) + \frac{\hat{G}_{y^5}(0, 0)}{20}\hat{E}_{y^3}(0, 0), \tag{3.28}$$

$$\kappa^{\mathrm{int}}_{3,5}(p) := -\hat{E}_{y^5}(0, 0) + \frac{\hat{G}_{y^5}(0, 0)}{12}\hat{E}_{y^4}(0, 0) - \left(\frac{7\hat{G}_{y^5}(0, 0)^2}{1200} - \frac{\hat{G}_{y^6}(0, 0)}{30}\right)\hat{E}_{y^3}(0, 0). \tag{3.29}$$

Similarly, evaluating the above expressions at each point $(x, 0)$ along the singular curve, we obtain smooth functions $\kappa^{\mathrm{int}}_{3,4}(x)$ and $\kappa^{\mathrm{int}}_{3,5}(x)$ along the singular curve.

Proposition 3.10. *Both $\kappa^{\mathrm{int}}_{3,4}$ and $\kappa^{\mathrm{int}}_{3,5}$ defined in Definition 3.9 are intrinsic invariants. Under changes of NACS, $\kappa^{\mathrm{int}}_{3,4}$ is unchanged, while $\kappa^{\mathrm{int}}_{3,5}$ changes sign when the null direction is reversed. In particular, the conditions $\kappa^{\mathrm{int}}_{3,5} = 0$ and $\kappa^{\mathrm{int}}_{3,5} \neq 0$ are intrinsic.*

Proof. Let $\tilde{\kappa}^{\mathrm{int}}_{3,4}, \tilde{\kappa}^{\mathrm{int}}_{3,5}$ denote the quantities computed in another NACS $(\tilde{x}, \tilde{y})$. Substituting (3.19), (3.20), and (3.22) of Lemma 3.6 into (3.28), we obtain

$$\tilde{\kappa}^{\mathrm{int}}_{3,4} = -\hat{E}_{y^4} - 6y_{\tilde{y}^2}\hat{E}_{y^3} + \frac{y_{\tilde{y}}\left(\hat{G}_{y^5} + 120y_{\tilde{y}^2}\right)}{20} y_{\tilde{y}}\hat{E}_{y^3} = -\hat{E}_{y^4} + \frac{\hat{G}_{y^5}}{20}\hat{E}_{y^3} = \kappa^{\mathrm{int}}_{3,4}.$$

Similarly, substituting (3.19)–(3.23) into (3.29), all terms involving $y_{\tilde{y}^2}(\tilde{x}, 0)$ and $y_{\tilde{y}^3}(\tilde{x}, 0)$ cancel, and we obtain $\tilde{\kappa}^{\mathrm{int}}_{3,5} = y_{\tilde{y}}\kappa^{\mathrm{int}}_{3,5}$ $(y_{\tilde{y}} = \pm 1)$. □

3.4 Shape of $(3, 4)$-cuspidal edges

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In this subsection, we define convexity and saddle-shapedness to describe the shape of $(3, 4)$-cuspidal edges and prove Proposition 3.12.

Definition 3.11. Let $f : \Sigma \to \mathbb{R}^3$ be a frontal with a 3-type edge at $p \in \Sigma$. We define the limiting tangent plane at p by

$$LT_p := f(p) + \nu(p)^{\perp}.$$

We say that f is *convex* at p if there exists a neighborhood $V \subset \Sigma$ of p such that

$$f(V) \cap LT_p = \{f(p)\}$$

and $f(V) \setminus \{f(p)\}$ lies entirely on one side of LT_p. On the other hand, we say that f is *saddle-shaped* at p if, for every neighborhood V of p, $f(V)$ meets both sides of LT_p.

Proposition 3.12. *Let $f : \Sigma \to \mathbb{R}^3$ be a frontal with a $(3, 4)$-cuspidal edge at $p \in \Sigma$. Then the following statements hold:*

(1) *If $\kappa^{\mathrm{int}}_{3,4}(p) > 0$, then f is convex at p.*

(2) *If $\kappa^{\mathrm{int}}_{3,4}(p) < 0$, then f is saddle-shaped at p.*

To prove this proposition, we prepare the following lemma.

Lemma 3.13. *Let $f : \Sigma \to \mathbb{R}^3$ be a frontal with a 3-type edge at $p \in \Sigma$. Let (u, v) be a local coordinate system near p satisfying*

$$S(f) = \{v = 0\}, \qquad f_v(u, 0) = f_{vv}(u, 0) = \mathbf{0}.$$

Then, along the singular curve, we have

$$\kappa^{\mathrm{int}}_{3,4}(u) = 2\frac{\det(f_u, f_{v^3}, f_{uu})\det(f_u, f_{v^3}, f_{v^4})}{\|f_u\|^{2/3}\|f_u \times f_{v^3}\|^{10/3}}(u). \tag{3.30}$$

Proof. First, we prove the formula in a NACS (x, y). Let $ds^2_f = \hat{E}\,dx^2 + 2\hat{F}\,dx\,dy + \hat{G}\,dy^2$ be the first fundamental form in this coordinate system. By the conditions of a NACS, we have

$$f_y = f_{yy} = \mathbf{0}, \qquad f_{xy} = f_{xyy} = \mathbf{0}$$

along the singular curve, and $\hat{F} \equiv 0$ also holds. Using these relations, we obtain

$$\hat{E}_{y^3} = -2\langle f_{xx}, f_{y^3}\rangle, \quad \hat{E}_{y^4} = -2\langle f_{xx}, f_{y^4}\rangle, \quad \hat{G}_{y^5} = 20\langle f_{y^3}, f_{y^4}\rangle.$$

Therefore, by the definition of $\kappa^{\mathrm{int}}_{3,4}$, we have

$$\kappa^{\mathrm{int}}_{3,4} = 2\left\{\langle f_{xx}, f_{y^4}\rangle - \langle f_{xx}, f_{y^3}\rangle\langle f_{y^3}, f_{y^4}\rangle\right\}. \tag{3.31}$$

On the other hand, by the definition of a NACS, we have $\|f_x\| = \|f_{y^3}\| = 1$, $\langle f_x, f_{y^3}\rangle = 0$ along the singular curve. Moreover, $\hat{E}(x, 0) = 1$ and $\hat{F} \equiv 0$ imply $\langle f_x, f_{xx}\rangle = \langle f_x, f_{y^4}\rangle = 0$. Since f_x and f_{y^3} form an orthonormal pair, we obtain

$$\langle f_{xx}, f_{y^4}\rangle - \langle f_{xx}, f_{y^3}\rangle\langle f_{y^3}, f_{y^4}\rangle = \det(f_x, f_{y^3}, f_{xx})\det(f_x, f_{y^3}, f_{y^4}).$$

Substituting this into (3.31), we obtain

$$\kappa_{3,4}^{\mathrm{int}} = 2\det(f_x, f_{y^3}, f_{xx})\det(f_x, f_{y^3}, f_{y^4}). \tag{3.32}$$

Next, let (u, v) be a general local coordinate system satisfying the assumptions of the lemma, and let (x, y) be a NACS. Write the coordinate transformation between them as

$$u = u(x, y), \qquad v = v(x, y).$$

Since the singular sets are given by $\{v = 0\}$ and $\{y = 0\}$, respectively, we have $v_x = 0$ along the singular curve. Moreover, by the chain rule, $f_{yy} = f_{vv} = \mathbf{0}$ and $f_{yy} = u_{yy} f_u = \mathbf{0}$ imply

$$v_x = 0, \qquad u_y = u_{yy} = 0 \quad \text{on } S(f).$$

Using these relations and the chain rule, we obtain

$$\begin{aligned} f_x &= u_x f_u, \\ f_x \times f_{y^3} &= u_x v_y^3 f_u \times f_{v^3}, \\ \det(f_x, f_{y^3}, f_{xx}) &= u_x^3 v_y^3 \det(f_u, f_{v^3}, f_{uu}), \qquad (3.33) \\ \det(f_x, f_{y^3}, f_{y^4}) &= u_x v_y^7 \det(f_u, f_{v^3}, f_{v^4}). \qquad (3.34) \end{aligned}$$

Since (x, y) is a NACS, we have $\|f_x\| = \|f_x \times f_{y^3}\| = 1$. Hence,

$$|u_x| = \frac{1}{\|f_u\|}, \qquad |v_y|^3 = \frac{\|f_u\|}{\|f_u \times f_{v^3}\|}. \tag{3.35}$$

Substituting (3.33)–(3.35) into (3.32), we obtain

$$\kappa_{3,4}^{\mathrm{int}} = 2u_x^4 v_y^{10} \det(f_u, f_{v^3}, f_{uu}) \det(f_u, f_{v^3}, f_{v^4}) = 2\frac{\det(f_u, f_{v^3}, f_{uu}) \det(f_u, f_{v^3}, f_{v^4})}{\|f_u\|^{2/3} \|f_u \times f_{v^3}\|^{10/3}}.$$

This proves (3.30). □

Proof of Proposition 3.12. Since a $(3, 4)$-cuspidal edge is also a $(3, 4)$-type edge in the sense of [4], we can apply [4, Proposition 2.9, (2.5)]. By an orientation-preserving change of coordinates in the source, together with a translation and a special orthogonal transformation in the target, we may assume that $p = (0, 0)$, $f(p) = \mathbf{0}$, and the limiting tangent plane at p is $z = 0$, and write

$$f(u, v) = \left(u,\ \frac{u^2 a(u)}{2} + \frac{v^3}{6},\ \frac{u^2 b_0(u)}{2} + \frac{v^4 b_4(u, v)}{24}\right), \qquad b_4(0, 0) \neq 0.$$

In this normal form, we have

$$\begin{aligned} f_u(p) &= (1, 0, 0), & f_{v^3}(p) &= (0, 1, 0), \\ f_{uu}(p) &= (0, a(0), b_0(0)), & f_{v^4}(p) &= (0, 0, b_4(0, 0)). \end{aligned}$$

Therefore,

$$\begin{gathered} \|f_u(p)\| = 1, \quad \|f_u(p) \times f_{v^3}(p)\| = 1, \\ \det(f_u, f_{v^3}, f_{uu})(p) = b_0(0), \quad \det(f_u, f_{v^3}, f_{v^4})(p) = b_4(0, 0). \end{gathered}$$

Thus, by the expression for $\kappa_{3,4}^{\mathrm{int}}$ in Lemma 3.13, we obtain

$$\kappa_{3,4}^{\mathrm{int}}(p) = 2b_0(0)b_4(0, 0). \tag{3.36}$$

On the other hand, since the limiting tangent plane at p is $z = 0$, the signed height function relative to this plane is given by

$$h(u, v) := \frac{u^2 b_0(u)}{2} + \frac{v^4 b_4(u, v)}{24}.$$

First, suppose that $\kappa_{3,4}^{\mathrm{int}}(p) > 0$. By (3.36), both $b_0(0)$ and $b_4(0, 0)$ are nonzero and have the same sign. Thus, after shrinking the neighborhood sufficiently, $b_0(u)$ and $b_4(u, v)$

have the same constant sign. Hence, $h(u,v)$ is nonzero and has a constant sign for $(u,v) \neq (0,0)$. Therefore, the image of f intersects the limiting tangent plane only at $f(p)$ and lies on one side of the plane. Thus, f is convex at p.

Next, suppose that $\kappa^{\mathrm{int}}_{3,4}(p) < 0$. By (3.36), both $b_0(0)$ and $b_4(0,0)$ are nonzero and have opposite signs. The signs of $b_0(u)$ and $b_4(u,v)$ are constant in a sufficiently small neighborhood. Now,

$$h(u,0) = \frac{u^2 b_0(u)}{2}, \qquad h(0,v) = \frac{v^4 b_4(0,v)}{24}$$

have opposite signs for sufficiently small nonzero u and v. Therefore, the image of f meets both sides of the limiting tangent plane in every neighborhood of p. Thus, f is saddle-shaped at p. □

仁知に、同じ文章を入れる。

Under the genericity assumption, the sufficient conditions in Proposition 3.12 become necessary and sufficient (Corollary 5.18).

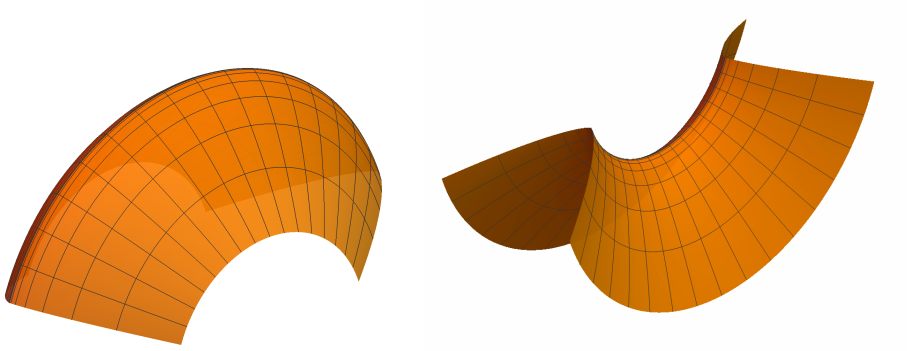


Figure 3: Local shapes of $(3,4)$-cuspidal edges (left: $\kappa^{\mathrm{int}}_{3,4} > 0$; right: $\kappa^{\mathrm{int}}_{3,4} < 0$)

Remark 3.14. For cuspidal edges, the relationship between the sign of the singular curvature κ_s and the local shape is known [7, Corollary 1.18]. In contrast, Proposition 3.12 determines the local shape of a $(3,4)$-cuspidal edge by the sign of the intrinsic invariant $\kappa^{\mathrm{int}}_{3,4}$.

3.5 Boundedness of Gaussian curvature near 3-type edges

(proof of theorem B) を追加する。

In this subsection, we derive the expansion of v^2K near the singular curve and characterize the boundedness of the Gaussian curvature K in terms of $\kappa^{\mathrm{int}}_{3,4}$ and $\kappa^{\mathrm{int}}_{3,5}$. Let (u,v) be a NACS, and let K denote the Gaussian curvature of f at regular points. When $F = 0$, Gauss's Theorema Egregium gives

$$K = \frac{E_v G_v + G_u^2}{4EG^2} + \frac{E_u G_u + E_v^2}{4E^2 G} - \frac{E_{vv} + G_{uu}}{2EG}. \tag{3.37}$$

Theorem 3.15. *Let $f : \Sigma \to \mathbb{R}^3$ be a frontal with a 3-type edge at $p \in \Sigma$. Then the following conditions are equivalent:*

(1) *There exists a neighborhood U of p such that K is bounded on $U \setminus S(f)$.*

(2) *There exists a neighborhood U of p such that $\kappa^{\mathrm{int}}_{3,4}$ and $\kappa^{\mathrm{int}}_{3,5}$ vanish identically on $S(f) \cap U$.*

To prove this theorem, we prepare the following lemma. Set

$$\widetilde{K}(u,v) := v^2 K(u,v).$$

Lemma 3.16. *In a NACS (x,y), we have*

$$\widetilde{K}(x,y) = \frac{1}{3}\kappa^{\mathrm{int}}_{3,4}(x) + \left(\frac{1}{6}\kappa^{\mathrm{int}}_{3,5}(x) - \frac{\hat{G}_{y^5}(x,0)}{360}\kappa^{\mathrm{int}}_{3,4}(x)\right) y + O(y^2). \tag{3.38}$$

Proof. By (3.17) and (3.18) of Corollary 3.5, we have

$$\hat{E}_y = O(y^2), \quad \hat{E}_{yy} = O(y), \quad \hat{E}_x = O(y^3), \quad \hat{G} = O(y^4), \quad \hat{G}_x = \hat{G}_{xx} = O(y^5).$$

Thus, after multiplying (3.37) by y^2, only

$$\frac{y^2\hat{E}_y\hat{G}_y}{4\hat{E}\hat{G}^2}, \qquad -\frac{y^2\hat{E}_{yy}}{2\hat{E}\hat{G}}$$

contribute to the constant and linear terms, while all other terms are $O(y^2)$.

For simplicity, for $j \geq 0$, we set $\hat{E}^0_{y^j}(x) := \hat{E}_{y^j}(x,0)$, $\hat{G}^0_{y^j}(x) := \hat{G}_{y^j}(x,0)$. Expanding the necessary coefficients of the first fundamental form in Taylor series, Corollary 3.5 yields

$$\begin{aligned}
\hat{E}_y(x,y) &= \frac{1}{2}\hat{E}^0_{y^3}(x)y^2 + \frac{1}{6}\hat{E}^0_{y^4}(x)y^3 + \frac{1}{24}\hat{E}^0_{y^5}(x)y^4 + O(y^5),\\
\hat{E}_{yy}(x,y) &= \hat{E}^0_{y^3}(x)y + \frac{1}{2}\hat{E}^0_{y^4}(x)y^2 + \frac{1}{6}\hat{E}^0_{y^5}(x)y^3 + O(y^4),\\
\hat{G}_y(x,y) &= y^3 + \frac{1}{24}\hat{G}^0_{y^5}(x)y^4 + \frac{1}{120}\hat{G}^0_{y^6}(x)y^5 + O(y^6),\\
\frac{1}{\hat{G}(x,y)} &= 4y^{-4}\left\{1 - \frac{\hat{G}^0_{y^5}(x)}{30}y + \left(\frac{\hat{G}^0_{y^5}(x)^2}{900} - \frac{\hat{G}^0_{y^6}(x)}{180}\right)y^2 + O(y^3)\right\},\\
\frac{1}{\hat{G}(x,y)^2} &= 16y^{-8}\left\{1 - \frac{\hat{G}^0_{y^5}(x)}{15}y + \left(\frac{\hat{G}^0_{y^5}(x)^2}{300} - \frac{\hat{G}^0_{y^6}(x)}{90}\right)y^2 + O(y^3)\right\},\\
\frac{1}{\hat{E}(x,y)} &= 1 + O(y^3).
\end{aligned}$$

Substituting these expansions, the terms of order y^{-1} from the two expressions cancel, and we obtain

$$\begin{aligned}
\widetilde{K}(x,y) = y^2K(x,y) = &-\frac{1}{3}\hat{E}^0_{y^4}(x) + \frac{\hat{G}^0_{y^5}(x)}{60}\hat{E}^0_{y^3}(x)\\
&+\left\{-\frac{1}{6}\hat{E}^0_{y^5}(x) + \frac{\hat{G}^0_{y^5}(x)}{60}\hat{E}^0_{y^4}(x) + \left(\frac{\hat{G}^0_{y^6}(x)}{180} - \frac{\hat{G}^0_{y^5}(x)^2}{900}\right)\hat{E}^0_{y^3}(x)\right\}y + O(y^2).
\end{aligned}$$

Finally, using (3.28) and (3.29) of Definition 3.9, the constant term is $\kappa^{\mathrm{int}}_{3,4}/3$, and the coefficient of y is $\kappa^{\mathrm{int}}_{3,5}/6 - \hat{G}_{y^5}\kappa^{\mathrm{int}}_{3,4}/360$. Thus, we obtain (3.38). □

By the above lemma, to determine the boundedness of K, it suffices to examine the vanishing of the constant and linear terms in the expansion of y^2K.

Proof of Theorem 3.15. By shrinking the neighborhood U of p if necessary, we may assume that a NACS (x,y) is defined on U and that $S(f)\cap U = \{y=0\}$. The remainder term in Lemma 3.16 can be expressed using a smooth function $k(x,y)$ as

$$\widetilde{K}(x,y) = \frac{1}{3}\kappa^{\mathrm{int}}_{3,4}(x) + \left(\frac{1}{6}\kappa^{\mathrm{int}}_{3,5}(x) - \frac{\hat{G}_{y^5}(x,0)}{360}\kappa^{\mathrm{int}}_{3,4}(x)\right)y + y^2k(x,y).$$

First, suppose that $\kappa^{\mathrm{int}}_{3,4} \equiv \kappa^{\mathrm{int}}_{3,5} \equiv 0$ on $S(f)\cap U$. Then,

$$y^2K(x,y) = y^2k(x,y).$$

Hence, $K(x,y) = k(x,y)$ at regular points. By shrinking U further if necessary, k is bounded on U. Therefore, K is bounded on $U\setminus S(f)$.

Conversely, suppose that K is bounded on $U\setminus S(f)$. For any $(x_0,0)\in S(f)\cap U$, we have

$$y^2K(x_0,y) \longrightarrow 0 \qquad (y\longrightarrow 0).$$

Letting $y \to 0$ in the above expansion of $\widetilde{K}$, we obtain

$$0 = \frac{1}{3}\kappa^{\mathrm{int}}_{3,4}(x_0).$$

Since x_0 is arbitrary, we have $\kappa^{\mathrm{int}}_{3,4} \equiv 0$ on $S(f) \cap U$. Dividing the expansion of $\widetilde{K}$ by y, we then obtain

$$yK(x,y) = \frac{1}{6}\kappa^{\mathrm{int}}_{3,5}(x) + yk(x,y).$$

Fixing $x = x_0$ and letting $y \to 0$ again, the boundedness of K implies that the left-hand side tends to 0, and hence $\kappa^{\mathrm{int}}_{3,5}(x_0) = 0$. Since x_0 is arbitrary, we conclude that $\kappa^{\mathrm{int}}_{3,5} \equiv 0$ on $S(f) \cap U$. □

Remark 3.17. For non-degenerate singular points of the first kind, [6, Theorem 3.9] shows that K is bounded $\iff$ product curvature $\kappa_\Pi \equiv 0$. In contrast, our theorem shows that, for 3-type edges, the boundedness of K is characterized by the simultaneous vanishing of the two intrinsic invariants $\kappa^{\mathrm{int}}_{3,4}$ and $\kappa^{\mathrm{int}}_{3,5}$ along the singular curve.

Example 3.18. Both of the following maps have $(3,4)$-cuspidal edges along the singular set $\{y = 0\}$, and the given coordinates satisfy the conditions of a NACS.

(1)
$$f_0(x,y) = \left(x, \frac{y^3}{6}, \frac{y^4}{24}\right).$$
In this case, $\kappa^{\mathrm{int}}_{3,4} = \kappa^{\mathrm{int}}_{3,5} = 0$, and the Gaussian curvature at regular points is $K = 0$.

(2)
$$f_{-1}(x,y) = \left(x - \frac{y^7}{1008}, \frac{y^3}{6}\cos x - \frac{y^4}{24}\sin x, \frac{y^3}{6}\sin x + \frac{y^4}{24}\cos x\right).$$
In this case, we also have $\kappa^{\mathrm{int}}_{3,4} = \kappa^{\mathrm{int}}_{3,5} = 0$, and $K \to -1$ as $y \to 0$.

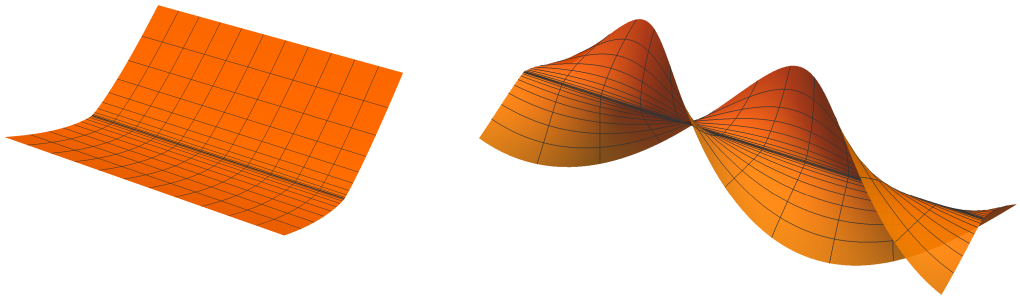


Figure 4: Examples of surfaces with bounded Gaussian curvature (left: $K = 0$; right: $K \to -1$ near the singular set)

4 $(3,4)$-カスプ辺および $(3,5)$-カスプ辺の判定条件

本節では，まず退化次数 n 特異点を導入し，第一種退化次数 $m-1$ 特異点と m-type edge の同値性を示す．次に，$(3,4)$-カスプ辺および $(3,5)$-カスプ辺の判定条件を与える．とくに，$(3,4)$-カスプ辺については，退化次数 2 特異点を用いた判定条件を導く．これらの判定条件は，第 5 節において内在的判定に用いる．

4.1 退化次数 n 特異点の導入

$f : \Sigma \to \mathbb{R}^3$ をフロンタル，$p \in S(f)$ を階数 1 の特異点とする．p の座標近傍 $(U; u, v)$ と，f に沿う局所単位法ベクトル場 $\nu : U \to S^2$ を取り，符号付き面積密度関数を

$$\lambda(u,v) := \det(f_u(u,v), f_v(u,v), \nu(u,v))$$

と定める．

Definition 4.1. フロンタル $f : \Sigma \to \mathbb{R}^3$ の特異点 $p \in S(f)$ が $d\lambda(p) \neq 0$ を満たすとき，p を非退化特異点という．

Definition 4.2 (退化次数 n 特異点). $n \geq 1$ を整数とする．rank 1 特異点 $p \in S(f)$ が**退化次数 n 特異点**であるとは，p の近傍で定義された滑らかな実数値関数 $\widehat{\lambda}, \alpha$ が存在して

$$\lambda = \alpha \widehat{\lambda}^n, \qquad d\widehat{\lambda}(p) \neq 0, \qquad \alpha(p) \neq 0$$

を満たすことをいう．

特に $n = 1$ のとき，退化次数 1 特異点は非退化特異点と一致する．退化次数 n 特異点 p に対して，必要に応じて近傍 U を縮小すれば，$\alpha \neq 0$ かつ $d\widehat{\lambda} \neq 0$ が U 上で成り立つ．したがって，

$$S(f) \cap U = \{\lambda = 0\} = \{\widehat{\lambda} = 0\}$$

であるから，陰関数定理より $S(f) \cap U$ は正則曲線となる．

次に，退化次数が定義域および値域の座標変換に依存しないことを確認する．

Proposition 4.3 (退化次数の $\mathcal{A}$-同値不変性)**.** 退化次数 n 特異点は $\mathcal{A}$-同値のもとで保たれる．つまり，定義域と値域の座標変換に依らない概念である．

証明は付録 A を参照．

以下では，第一種性と退化次数 n 特異点が内在的な性質であることを確認する．実際，特異点集合 $S(f)$ と退化方向 $\ker(df)_p$ は第一基本形式のみから決まるため，第一種性は内在的な性質である．なお，非退化特異点に対する同様の内在的な特徴付けは [9, Definition 5.2] を参照．一方，一般の局所座標 (u, v) において $\lambda^2 = EG - F^2$ が成り立つ．退化次数 n 特異点では，適当な局所座標系 (u, v) により

$$\lambda = v^n \alpha, \qquad \alpha(p) \neq 0$$

と書けるから，

$$EG - F^2 = v^{2n} \alpha^2$$

を得る．したがって，第一種性と退化次数 n 特異点は内在的な性質である．

次に，m-type edge が内在的に特徴付けられることを命題 4.4 で示す．ここで，rank 1 特異点 p が第一種であることの定義 (定義 2.4) を再掲すると，$S(f)$ が p の近傍で正則曲線であり，$T_p S(f) \neq \ker(df_p)$ を満たすことである．

Proposition 4.4. $m \geq 2$ を整数，$f : \Sigma \to \mathbb{R}^3$ をフロンタル，$p \in S(f)$ を *rank* 1 特異点とする．このとき，次の条件は同値である．

(1) p は第一種かつ退化次数 $m - 1$ 特異点である．

(2) f は p で *m-type edge* である．

Proof. まず，p が第一種かつ退化次数 $m - 1$ 特異点であると仮定する．第一種性より，特異方向と退化方向は特異曲線上で横断的である．したがって，特異曲線を u 軸とし，その上で ∂_v が退化方向を与える局所座標系 (u, v) を取ることができる．このとき，

$$S(f) = \{v = 0\}, \qquad f_v(u, 0) = \mathbf{0}, \qquad f_u(u, 0) \neq \mathbf{0}$$

が成り立つ．この座標系に関する第一基本形式を $ds_f^2 = E\,du^2 + 2F\,du\,dv + G\,dv^2$ と書く．必要に応じて近傍を縮小すれば，$E > 0$ としてよい．ここで，ベクトル場

$$\xi := \partial_u, \qquad \eta := \partial_v - \frac{F}{E}\partial_u$$

を定める．特異曲線上では $F(u, 0) = 0$ であるから，η は退化方向を与える．また，$\langle f_u, \eta f \rangle = 0$ であり，

$$\|\eta f\|^2 = G - \frac{F^2}{E} = \frac{EG - F^2}{E} = \frac{\lambda^2}{E}$$

が成り立つ．退化次数 $m - 1$ 特異点の定義より，

$$\lambda = \alpha \widehat{\lambda}^{m-1}, \qquad d\widehat{\lambda}(p) \neq 0, \qquad \alpha(p) \neq 0$$

と書ける．特異点集合が u 軸であることから $\widehat{\lambda}(u,0)=0$ であり，$d\widehat{\lambda}(p)\neq 0$ より $\widehat{\lambda}_v(p)\neq 0$ を得る．したがって，因子の補題により $\rho(p)\neq 0$ を満たす滑らかな関数 ρ を用いて

$$\widehat{\lambda}(u,v)=v\rho(u,v)$$

と書ける．よって，

$$\|\eta f\|^2=v^{2m-2}\frac{\alpha^2\rho^{2m-2}}{E} \tag{4.1}$$

が成り立つ．式 (4.1) より $\|\eta f\|=O(v^{m-1})$ であるから，

$$(\eta f)_{v^j}(u,0)=\mathbf{0}\qquad (j=0,\dots,m-2)$$

が従う．したがって，因子の補題により滑らかな $\mathbb{R}^3$ 値写像 ψ を用いて

$$\eta f=v^{m-1}\psi \tag{4.2}$$

と書ける．式 (4.1) より $\|\psi\|^2=\alpha^2\rho^{2m-2}/E$ であるから，$\psi(p)\neq\mathbf{0}$ である．また，$\langle f_u,\eta f\rangle=0$ より $\langle f_u,\psi\rangle=0$ が成り立つ．一方，$\eta v=1$ であるから，式 (4.2) を η で繰り返し微分すると，

$$\eta^j f=\mathbf{0}\quad (j=1,\dots,m-1)\quad \text{on } S(f),\qquad \eta^m f(p)=(m-1)!\psi(p)$$

を得る．$f_u(p)\neq\mathbf{0}$，$\psi(p)\neq\mathbf{0}$，$\langle f_u(p),\psi(p)\rangle=0$ であるから，

$$\operatorname{rank}\big(\xi f(p),\eta^m f(p)\big)=2$$

を得る．よって，[4, Proposition 2.2] を適用すると，f は p で m-type edge である．

逆に，f が p で m-type edge であると仮定する．定義より，f は

$$g(u,v)=(u,v^m,v^{m+1}a(u,v))$$

と $\mathcal{A}$-同値である．g の特異点集合は $\{v=0\}$ であり，その特異方向と退化方向はそれぞれ ∂_u，∂_v であるから，原点は第一種特異点である．さらに，第一種性は $\mathcal{A}$-同値のもとで保存される．また，直接計算より

$$g_u\times g_v=v^{m-1}\psi,\qquad \psi(0,0)=(0,0,m)$$

を満たす滑らかな写像 ψ が存在する．したがって，単位法ベクトル場を $\nu=\psi/\|\psi\|$ と取れば，g の符号付き面積密度関数は

$$\lambda=v^{m-1}\|\psi\|,\qquad \|\psi(0,0)\|=m\neq 0$$

となる．よって，原点は退化次数 $m-1$ 特異点である．命題 4.3 より，退化次数 $m-1$ 特異点であることは $\mathcal{A}$-同値のもとで保存される．すでに第一種性も示したので，主張が示される． □

Remark 4.5. 退化次数 n 特異点は必ずしも第一種とは限らず，第二種の例も存在する．したがって，退化次数 n 特異点は $(n+1)$-type edge を含む，より広いクラスである．図 5 に，$n=2$ の場合として第一種特異点と第二種特異点の例を示す．

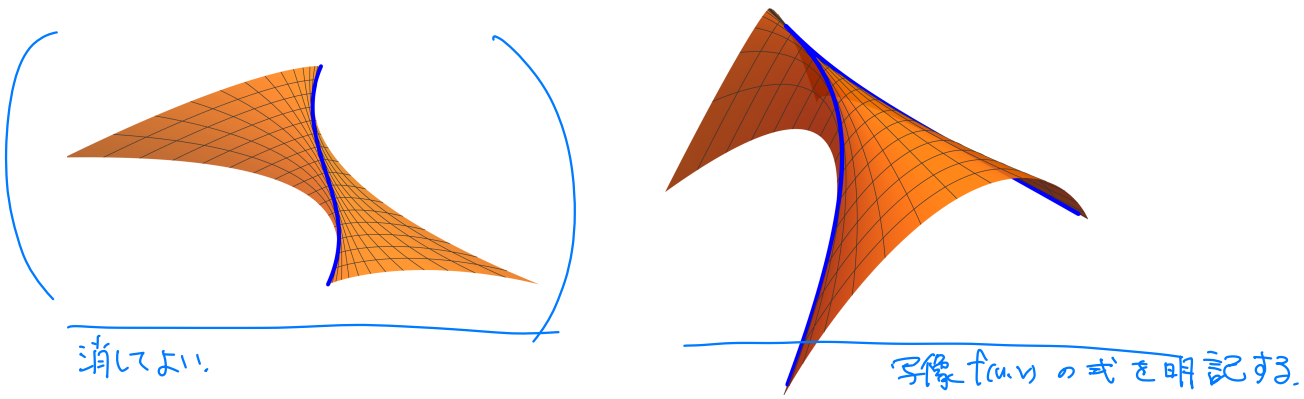


Figure 5: 退化次数 2 特異点の例．左図：第一種特異点，右図：原点で第一種ではない特異点．(青色部分は特異点集合の像を表す)

• prop 4.4 を用いて，(3,4)-CE の判定条件を内在的に述べる．

4.2 $(3,4)$-カスプ辺の判定条件

Theorem 4.6 ($(3,4)$-カスプ辺の判定条件)**.** $f:\Sigma\to\mathbb{R}^3$ を $p\in\Sigma$ でフロント（波面）とする．このとき，次の条件は同値である．

(1) f は p で $(3,4)$-カスプ辺である．

(2) f は p で 3-*type edge* である．

この証明のために，ホイットニーの補題 (事実 4.7) と補題 4.8 を準備する．

Fact 4.7 (ホイットニーの補題 [11])**.** $f(u,v)$ を，$\mathbb{R}^2$ の原点付近で定義された C^∞ 級関数で，$f(u,v)=f(u,-v)$ が成り立つとすると，$\mathbb{R}^2$ の原点付近で定義された C^∞ 級関数 $g(u,v)$ が存在し，原点付近で $f(u,v)=g(u,v^2)$ と書ける．一方，$f(u,v)=-f(u,-v)$ が成り立つとすると，$\mathbb{R}^2$ の原点付近で定義された C^∞ 級関数 $h(u,v)$ が存在し，原点付近で $f(u,v)=vh(u,v^2)$ と書ける．

この補題より，任意の滑らかな関数 $C(u,v)$ は，滑らかな関数 C_0,C_1 を用いて

$$C(u,v)=C_0(v^2,u)+vC_1(v^2,u)$$

と表せる．

Lemma 4.8. $f:\Sigma\to\mathbb{R}^3$ をフロンタル，$p\in\Sigma$ とし，f は p で 3-*type edge* であるとする．このとき，$p=(0,0)$，$f(p)=\mathbf{0}$ となるように定義域と値域の局所座標を適当に取れば，ある滑らかな関数 $b_1(u),b_2(u),b_3(u,v)$ を用いて

$$f(u,v)=\bigl(v^3,\ v^4b_1(u)+v^5b_2(u)+v^6b_3(u,v),\ u\bigr)$$

と表すことができる．

Proof. f は p で 3-type edge であるから，定義 2.2 より定義域と値域の座標変換を用いて

$$f(u,v)=(v^3,v^4a(u,v),u)$$

と表せる．ここで，$a(u,v)$ を $v=0$ のまわりで Taylor 展開すると，滑らかな関数 $b_1(u),b_2(u),b_3(u,v)$ が存在して

$$a(u,v)=b_1(u)+vb_2(u)+v^2b_3(u,v)$$

と書ける．これを代入すれば主張を得る． □

定理 *4.6* の証明．まず，$(3,4)$-カスプ辺は定義より 3-type edge であるから，(1) ⇒ (2) が成り立つ．

次に (2) ⇒ (1) を示す．p を原点とし，$f(p)=\mathbf{0}$ としてよい．補題 4.8 より，ある局所座標 (u,v) と滑らかな関数 $b_1(u),b_2(u),b_3(u,v)$ を用いて

$$f(u,v)=\bigl(v^3,\ v^4b_1(u)+v^5b_2(u)+v^6b_3(u,v),\ u\bigr) \tag{4.3}$$

と書ける．

また，f がフロントであることは $\mathcal{A}$-同値のもとで保存されるため $b_1(u)\neq 0$ が従う．実際，$f_v/v^2\times f_u$ を単位化した法ベクトル場 ν は，

$$\nu_v(u,0)=\left(\frac{4}{3}b_1(u),0,0\right)$$

を満たす．$f_v(u,0)=0$ より，ルジャンドルリフトのはめ込み性から $\nu_v(u,0)\neq\mathbf{0}$ が従う．したがって，$b_1(u)\neq 0$ である．値域の座標変換 $(X,Y,Z)\longmapsto(X,Y/b_1(Z),Z)$ により，式 (4.3) は

$$f(u,v)=\bigl(v^3,v^4+v^5c(u)+v^6d(u,v),u\bigr) \tag{4.4}$$

としてよい．ここで定義域の座標変換

$$v=\tilde{v}-\frac{c(u)}{4}\tilde{v}^2-\frac{c(u)^2}{16}\tilde{v}^3$$

を用い，続いて値域の座標変換

$$(X,Y,Z)\longmapsto\left(X+\frac{3c(Z)}{4}Y,\ Y,\ Z\right)$$

を施すと，滑らかな関数 A,B を用いて式 (4.4) は

$$f(u,v)=(v^3,v^4,u)+v^6(A(u,v),B(u,v),0) \tag{4.5}$$

と書ける．ここで A,B に対してホイットニーの補題を 2 回適用する．滑らかな関数 A_j,B_j $(j=0,1,2,3)$ を用いて

$$A(u,v)=A_0(v^4,u)+vA_1(v^4,u)+v^2A_2(v^4,u)+v^3A_3(v^4,u),$$
$$B(u,v)=B_0(v^4,u)+vB_1(v^4,u)+v^2B_2(v^4,u)+v^3B_3(v^4,u)$$

と書ける．$X=v^3$, $Y=v^4$, $Z=u$ と置くと

$$v^6=X^2,\qquad v^7=XY,\qquad v^8=Y^2,\qquad v^9=X^3$$

である．値域上の滑らかな関数 P,Q を

$$P(X,Y,Z):=X^2A_0(Y,Z)+XYA_1(Y,Z)+Y^2A_2(Y,Z)+X^3A_3(Y,Z),$$
$$Q(X,Y,Z):=X^2B_0(Y,Z)+XYB_1(Y,Z)+Y^2B_2(Y,Z)+X^3B_3(Y,Z)$$

と定めると

$$P(v^3,v^4,u)=v^6A(u,v),\qquad Q(v^3,v^4,u)=v^6B(u,v)$$

が成り立つ．したがって値域の座標変換を

$$\Phi(X,Y,Z):=(X+P(X,Y,Z),Y+Q(X,Y,Z),Z)$$

と置けば，式 (4.5) より，標準形 $f_0(u,v)=(v^3,v^4,u)$ に対して

$$\Phi\circ f_0=(v^3,v^4,u)+v^6(A(u,v),B(u,v),0)=f$$

となる．$(dP)_{(0,0,0)}=(dQ)_{(0,0,0)}=0$ より $\det(d\Phi_{(0,0,0)})=1\neq 0$ であることから，逆関数定理より Φ は原点近傍の局所微分同相である．したがって元の写像芽は f_0 と $\mathcal{A}$-同値であり，p は $(3,4)$-カスプ辺である． □

命題 4.4 と定理 4.6 より，$(3,4)$-カスプ辺の判定条件に関して次の系を得る．

Corollary 4.9 ($(3,4)$-カスプ辺の判定条件)**.** $f:\Sigma\to\mathbb{R}^3$ をフロント *(*波面*)*，$p\in S(f)$ を退化次数 2 特異点とする．このとき，次の条件は同値である．

(1) f は p で $(3,4)$-カスプ辺である．

(2) p は第一種特異点である．

Remark 4.10. フロントの非退化特異点では，カスプ辺であることと第一種であることが同値であることが知られている [1, Proposition 1.3]．系 4.9 は，これに対応する退化次数 2 の場合の判定条件である．

4.3 $(3,5)$-カスプ辺の判定条件

Theorem 4.11 ($(3,5)$-カスプ辺の判定条件)**.** $f:\Sigma\to\mathbb{R}^3$ をフロンタル，$p\in\Sigma$ とする．このとき，次の条件は同値である．

(1) f は p で $(3,5)$-カスプ辺である．

(2) f は p で 3-*type edge* であり，p の近傍 U が存在して，$S(f)\cap U$ の各点でフロントではなく，さらに

$$\det(\xi f,\eta^3 f,\eta^5 f)(p)\neq 0$$

が成り立つ．

ただし，(ξ,η) は事実 *2.6* の条件を満たす特異ベクトル場と退化ベクトル場の組とする．

条件 (2) では，非フロント性は p の一点だけでなく $S(f)\cap U$ の全点で仮定していることに注意する．証明は付録 B を参照．

5　特異点型の内在的判定と等長性による保存

本節では，前節までの結果を用いて定理 5.12 を証明し，その帰結として系 5.19 を導く．そのために，符号付き極限法曲率と補正されたカスプ的曲率を導入し，これらを用いて積曲率を定義する．さらに，積曲率と $\kappa_{3,4}^{\mathrm{int}}, \kappa_{3,5}^{\mathrm{int}}$ の関係を述べ，特異点型の内在的判定条件を与える．最後に，第 4.1 節で得た m-type edge の内在的特徴付けを用いて，特異点型が局所等長性によって保存されることを示す．

5.1　符号付き極限法曲率とジェネリック性

本 subsection では，3-type edge に対する符号付き極限法曲率を定義し，その不変性を示す．さらに，ジェネリックを導入し，符号付き極限法曲率との関係を述べる．

Definition 5.1 (正に向き付けられた). p の近傍 U に向きを固定する．ベクトル場の組 (ξ, η) が**正に向き付けられている**とは，任意の $q \in S(f) \cap U$ において，順序付き基底 (ξ_q, η_q) の向きが U に固定した向きと一致することをいう．

Definition 5.2 (符号付き極限法曲率). $f : \Sigma \to \mathbb{R}^3$ をフロンタルとし，f は $p \in \Sigma$ で 3-type edge とする．p の近傍 U に向きを固定し，(ξ, η) を事実 2.6 の条件を満たす正に向き付けられた特異ベクトル場と退化ベクトル場の組とする．p における符号付き極限法曲率を

$$\widetilde{\kappa}_\nu(p) := \frac{\det\big(\xi f(p), \eta^3 f(p), \xi^2 f(p)\big)}{\|\xi f(p) \times \eta^3 f(p)\| \, \|\xi f(p)\|^2}$$

によって定める．

Lemma 5.3 (符号付き極限法曲率の well-definedness). $f : \Sigma \to \mathbb{R}^3$ をフロンタル，$p \in \Sigma$ とし，f は p で *3-type edge* であるとする．このとき，$\widetilde{\kappa}_\nu$ は事実 *2.6* の条件を満たす正に向き付けられたベクトル場の組 (ξ, η) の取り方に依らない．

Proof. p の十分小さな開近傍 U において (ξ, η) に適合する局所座標 (u, v) を取り，

$$S(f) \cap U = \{v = 0\}, \qquad \xi = \partial_u \quad \text{on } S(f) \cap U, \qquad \eta = \partial_v \quad \text{on } U$$

とする．3-type edge であるから，ある滑らかな写像 ψ を用いて $f_v = v^2 \psi$ と書ける．

いま，条件を満たす正に向き付けられた別のベクトル場の組を $(\overline{\xi}, \overline{\eta})$ とする．特異曲線上で，ある非零の滑らかな関数 α を用いて

$$\overline{\xi} = \alpha(u)\, \partial_u$$

と書ける．また，[4, Lemma 2.6] の証明より，ある非零の滑らかな関数 h と滑らかな関数 a を用いて

$$\overline{\eta} = h(u, v) \left(v^2 a(u, v) \partial_u + \partial_v\right)$$

と書ける．$f_v = v^2 \psi$ より，特異曲線上で $\overline{\eta}^3 f = h^3 \big(f_{v^3} + 2a f_u\big)$ が成り立つ．一方，$\overline{\xi}^2 f$ も特異曲線上で $\overline{\xi}^2 f = \alpha^2 f_{uu} + \alpha \alpha' f_u$ が成り立つ．したがって，

$$\det\big(\overline{\xi} f, \overline{\eta}^3 f, \overline{\xi}^2 f\big) = \alpha^3 h^3 \det(f_u, f_{v^3}, f_{uu}),$$

および

$$\|\overline{\xi} f \times \overline{\eta}^3 f\| \, \|\overline{\xi} f\|^2 = |\alpha h^3| \alpha^2 \|f_u \times f_{v^3}\| \, \|f_u\|^2$$

を得る．よって

$$\widetilde{\kappa}_\nu(\overline{\xi}, \overline{\eta}) = \operatorname{sgn}(\alpha h)\, \widetilde{\kappa}_\nu(\xi, \eta)$$

が成り立つ．仮定より両方の組が正に向き付けられているため，特異曲線上では $\alpha h > 0$ であり，

$$\widetilde{\kappa}_\nu(\overline{\xi}, \overline{\eta}) = \widetilde{\kappa}_\nu(\xi, \eta)$$

が成り立つ．よって，$\widetilde{\kappa}_\nu$ はベクトル場の組の取り方によらず定まる． □

Remark 5.4 (極限法曲率との関係). $f : \Sigma \to \mathbb{R}^3$ をフロンタル，$p \in \Sigma$ を rank 1 特異点とし，ν を f に沿う単位法ベクトル場とする．また，$f_v(p) = \mathbf{0}$ を満たす局所座標系 (u, v) を取る．rank 1 であることより，$f_u(p) \neq \mathbf{0}$ である．このときの p における極限法曲率を

$$\kappa_\nu(p) := \frac{\langle f_{uu}(p), \nu(p) \rangle}{\|f_u(p)\|^2}$$

によって定める．この値は $f_v(p) = \mathbf{0}$ を満たす座標系の取り方によらない．$\kappa_\nu(p)$ を f の p における**極限法曲率**という [6][7]．これは ν の取り方に依存する．

3-type edge に対して，事実 2.6 の条件を満たす正に向き付けられたベクトル場の組 (ξ, η) を取り，特異曲線上で

$$\nu_{\xi,\eta} := \frac{\xi f \times \eta^3 f}{\|\xi f \times \eta^3 f\|}$$

と単位法ベクトル場の向きを取れば，

$$\widetilde{\kappa}_\nu(p) = \kappa_{\nu_{\xi,\eta}}(p)$$

が成り立つ．したがって，符号付き極限法曲率は，特異ベクトル場と退化ベクトル場の組によって単位法ベクトル場の向きを固定した極限法曲率とみなすことができる．とくに，符号付き極限法曲率は ν の選び方に依存しない概念である．

次に，符号付き極限法曲率の非零性によって特徴付けられるジェネリック性を導入する．

Definition 5.5. $f : \Sigma \to \mathbb{R}^3$ をフロンタル，ν を f に沿う単位法ベクトル場とする．特異点 $p \in S(f)$ の近傍で，特異点集合 $S(f)$ が正則曲線 $\gamma : (-\varepsilon, \varepsilon) \to \Sigma$ $(\varepsilon > 0)$ によってパラメータ付けられており，$\gamma(0) = p$ を満たすとする．このとき，p が**ジェネリック**であるとは，

- $\hat{\gamma}(t) := f \circ \gamma(t)$ とおくとき，$\hat{\gamma}'(0), \hat{\gamma}''(0)$ が一次独立であり，
- $f(p) + \mathrm{Span}(\hat{\gamma}'(0), \hat{\gamma}''(0))$ が極限接平面 LT_p と一致しない

ときをいう．

generic なカスプ辺の概念は [7] により導入されたが，この定義は，その拡張である．

Proposition 5.6 (ジェネリック)**.** $f : \Sigma \to \mathbb{R}^3$ をフロンタル，$p \in S(f)$ とし，f が点 p で 3-*type edge* であるとする．このとき，p がジェネリックであることと

$$\widetilde{\kappa}_\nu(p) \neq 0$$

は同値である．

Proof. [13, Proposition 5.2.9] の証明と同様に示せる． □

5.2 補正された $(3,5)$-カスプ的曲率の導入

本 subsection では，[4] で導入された $(3,4)$-カスプ的曲率と $(3,5)$-カスプ的曲率を紹介する．さらに，$(3,5)$-カスプ的曲率を一般の 3-type edge に対して扱えるよう，補正された $(3,5)$-カスプ的曲率を導入する．

Fact 5.7 (カスプ的曲率, [4])**.** フロンタル $f : \Sigma \to \mathbb{R}^3$ は，$p \in \Sigma$ において 3-*type edge* であるとする．特異ベクトル場 ξ と退化ベクトル場 η は正に向き付けられ，事実 *2.6* の条件を満たすとする．このとき，

$$\omega_{3,4}(p) := \frac{\|\xi f(p)\|^{4/3} \det\big(\xi f(p), \eta^3 f(p), \eta^4 f(p)\big)}{\|\xi f(p) \times \eta^3 f(p)\|^{7/3}}$$

は，このような組 (ξ, η) の取り方によらない *([4, Proposition 2.10])*. $\omega_{3,4}(p)$ を f の p における $(3,4)$-**カスプ的曲率**という．

さらに，p の近傍 U が存在して，$S(f) \cap U$ 上で $\omega_{3,4} \equiv 0$ が成り立つとする．$S(f) \cap U$ に沿う退化方向の向きを固定し，η はその向きと一致するとする．このとき，

$$\omega_{3,5}(p) := \frac{\|\xi f(p)\|^{5/3} \det\big(\xi f(p), \eta^3 f(p), \eta^5 f(p)\big)}{\|\xi f(p) \times \eta^3 f(p)\|^{8/3}} \tag{5.1}$$

は，η が固定した退化方向の向きと一致するような組 (ξ, η) の取り方によらない *([4, Proposition 2.13])*. $\omega_{3,5}(p)$ を f の p における $(3,5)$-**カスプ的曲率**という．

一方，$\omega_{3,4} \equiv 0$ を仮定しなければ，式 (5.1) の右辺は一般に退化ベクトル場の取り方に依存する．そこで，この依存性を打ち消す補正項を導入する．

$f : \Sigma \to \mathbb{R}^3$ をフロンタル，$p \in \Sigma$ とし，f は p で 3-type edge であるとする．$S(f)$ を f の特異点集合とする．p の十分小さい近傍 U を取り，U に向きを固定する．(ξ, η) を正に向き付けられたベクトル場の組とし，ξ, η は $S(f) \cap U$ 上で

$$\eta f = \eta^2 f = 0, \quad \operatorname{rank}(\xi f, \eta^3 f) = 2$$

を満たすものとする．以下，$S(f) \cap U$ 上で

$$X := \xi f, \qquad Y := \eta^3 f, \qquad Z := \eta^4 f, \qquad W := \eta^5 f$$

とおく．また，X に垂直な成分を

$$Y^\perp := Y - \frac{\langle X, Y \rangle}{\|X\|^2} X, \qquad Z^\perp := Z - \frac{\langle X, Z \rangle}{\|X\|^2} X$$

によって定める．$\operatorname{rank}(X, Y) = 2$ より $Y^\perp \neq 0$ である．ここで，特異曲線上の関数

$$\Omega_{3,5}(\xi, \eta) := \frac{\|X\|^{5/3} \det(X, Y, W)}{\|X \times Y\|^{8/3}} \tag{5.2}$$

を考える．この式は一般の 3-type edge に対して定義できるが，その値はベクトル場の組 (ξ, η) の取り方に依存する．

Definition 5.8 (補正された $(3,5)$-カスプ的曲率)．特異曲線上の関数

$$\Gamma(\xi, \eta) := \frac{\langle Y^\perp, Z^\perp \rangle}{\|Y^\perp\|^{7/3}} \tag{5.3}$$

を定める．このとき，

$$\widehat{\omega}_{3,5}(\xi, \eta) := \Omega_{3,5}(\xi, \eta) - \frac{5}{3} \Gamma(\xi, \eta)\, \omega_{3,4}(\xi, \eta) \tag{5.4}$$

を補正された $(3,5)$-カスプ的曲率とよぶ．

Proposition 5.9. 退化方向の向きを固定する．このとき，$\widehat{\omega}_{3,5}$ は事実 *2.6* の条件を満たす正に向き付けられたベクトル場の組 (ξ, η) の取り方に依らない．

退化方向の向きを反転すると $\widehat{\omega}_{3,5}$ の符号は反転する．特に，

$$|\widehat{\omega}_{3,5}|, \qquad \widehat{\omega}_{3,5} = 0, \qquad \widehat{\omega}_{3,5} \neq 0$$

は退化方向の向きによらない．

Appendix へ移動する.

Proof. まず，退化方向の向きを固定した場合を示す．以下の計算はすべて $S(f) \cap U$ 上で行う．ベクトル場の組の変更を，ξ の変更，η の ξ 方向成分の変更，η の正の関数倍に分けて調べる．

退化ベクトル場 η を固定する．$(\widetilde{\xi}, \eta)$ も正に向き付けられるような別の特異ベクトル場 $\widetilde{\xi}$ を取ると，$S(f) \cap U$ 上で $\widetilde{\xi} = c\xi$ と書ける．ここで $c > 0$ である．したがって，$\widetilde{\xi}$ に対応する X を $\widetilde{X}$ とすると，$\widetilde{X} = cX$ である．$\mathbb{R}X = \mathbb{R}\widetilde{X}$ であるから，$X^\perp$ への直交射影は変化せず，

$$\widetilde{Y}^\perp = Y^\perp, \qquad \widetilde{Z}^\perp = Z^\perp.$$

となる．式 (5.3) より，$\Gamma(\widetilde{\xi}, \eta) = \Gamma(\xi, \eta)$ である．また，$c > 0$ であるから，定義 (事実 5.7, 式 (5.2)) より

$$\frac{\|cX\|^{4/3} \det(cX, Y, Z)}{\|cX \times Y\|^{7/3}} = \omega_{3,4}, \qquad \frac{\|cX\|^{5/3} \det(cX, Y, W)}{\|cX \times Y\|^{8/3}} = \Omega_{3,5}(\xi, \eta)$$

である．Γ, $\omega_{3,4}$, $\Omega_{3,5}$ はいずれも不変であり，式 (5.4) より $\widehat{\omega}_{3,5}$ は ξ の取り方に依らない．

次に，η に ξ 方向成分を加える場合を考える．(u, v) を

$$S(f) \cap U = \{v = 0\}, \qquad \eta = \partial_v$$

となる適合座標系とする．前段で ξ の取り方によらないことを示したので，$\xi = \partial_u$ としてよい．特異点集合上の条件 $f_v = f_{vv} = 0$ より，ある滑らかなベクトル値関数 ψ を用いて

$$f_v = v^2\psi$$

と書ける．一方，[4, Lemma 2.6] の証明より，同じ条件を満たす別の退化ベクトル場は，非零関数倍を除けば

$$\bar{\eta} := \partial_v + v^2 a(u,v)\partial_u$$

の形に書ける．$\Phi := \psi + af_u$ とおけば $\bar{\eta}f = v^2\Phi$ であり，$\bar{\eta}v = 1$ より $S(f)\cap U$ 上で

$$\bar{\eta}^3 f = 2\Phi, \qquad \bar{\eta}^4 f = 6\bar{\eta}\Phi, \qquad \bar{\eta}^5 f = 12\bar{\eta}^2\Phi$$

となる．また $f_{uv} = f_{uvv} = 0$ なので，

$$\bar{\eta}\Phi = \psi_v + a_v f_u, \qquad \bar{\eta}^2\Phi = \psi_{vv} + a_{vv}f_u \quad \text{on } S(f)\cap U$$

が成り立つ．さらに $f_{v^3} = 2\psi$，$f_{v^4} = 6\psi_v$，$f_{v^5} = 12\psi_{vv}$ を用いると

$$\bar{\eta}^3 f = Y + 2aX, \qquad \bar{\eta}^4 f = Z + 6a_v X, \qquad \bar{\eta}^5 f = W + 12a_{vv}X$$

である．$X^\perp$ への射影を取れば $(\bar{\eta}^3 f)^\perp = Y^\perp$, $(\bar{\eta}^4 f)^\perp = Z^\perp$ である．ゆえに Γ についても，式 (5.3) より

$$\Gamma(\xi,\bar{\eta}) = \Gamma(\xi,\eta)$$

となる．また，上の三式 $\bar{\eta}^3 f, \bar{\eta}^4 f, \bar{\eta}^5 f$ で加えられる項はすべて $X = f_u$ の倍数であるから，$\det(X,Y,Z)$, $\det(X,Y,W)$ および $X\times Y$ は変化しない．したがって

$$\omega_{3,4}(\xi,\bar{\eta}) = \omega_{3,4}(\xi,\eta), \quad \Omega_{3,5}(\xi,\bar{\eta}) = \Omega_{3,5}(\xi,\eta).$$

となり，値は不変である．よって，式 (5.4) より

$$\widehat{\omega}_{3,5}(\xi,\bar{\eta}) = \widehat{\omega}_{3,5}(\xi,\eta)$$

となり，$\widehat{\omega}_{3,5}$ は η の ξ 方向成分の変更によって変化しない．

最後に η の正の関数倍を考える．$\widetilde{\eta} = h\eta$ $(h>0)$，$h_1 := \eta h$ とおく．また，$\widetilde{\eta}$ に対応する Y,Z,W をそれぞれ $\widetilde{Y},\widetilde{Z},\widetilde{W}$ と書く．$S(f)\cap U$ 上で $\eta f = \eta^2 f = 0$ であることを用いて微分すれば

$$\widetilde{Y} = h^3 Y, \quad \widetilde{Z} = h^4 Z + 6h^3 h_1 Y, \quad \widetilde{W} = h^5 W + 10h^4 h_1 Z + (25h^3 h_1^2 + 10h^4\eta^2 h)Y$$

を得る．これらを $\omega_{3,4}$，$\Omega_{3,5}$ の定義式に代入し，$\|Y^\perp\| = \|X\times Y\|/\|X\|$ を用いると

$$\omega_{3,4}(\xi,h\eta) = \omega_{3,4}(\xi,\eta), \qquad \Omega_{3,5}(\xi,h\eta) = \Omega_{3,5}(\xi,\eta) + \frac{10h_1}{h\|Y^\perp\|^{1/3}}\omega_{3,4}.$$

を得る．一方，直交射影を取ると

$$\widetilde{Y}^\perp = h^3 Y^\perp, \qquad \widetilde{Z}^\perp = h^4 Z^\perp + 6h^3 h_1 Y^\perp$$

であり，式 (5.3) から

$$\Gamma(\xi,h\eta) = \Gamma(\xi,\eta) + \frac{6h_1}{h\|Y^\perp\|^{1/3}}$$

となる．以上より，式 (5.4) から

$$\widehat{\omega}_{3,5}(\xi,h\eta) - \widehat{\omega}_{3,5}(\xi,\eta) = \left(10 - \frac{5}{3}\cdot 6\right)\frac{h_1}{h\|Y^\perp\|^{1/3}}\omega_{3,4} = 0$$

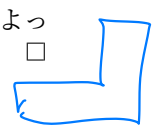
である．したがって，$\widehat{\omega}_{3,5}$ は η の正の関数倍によって変化しない．

[4, Lemma 2.6] の証明より，条件を満たす二つの退化ベクトル場の違いは，局所的には η の ξ 方向成分の変更と η の正の関数倍の組合せで記述できる．したがって $\widehat{\omega}_{3,5}$ は，退化方向の向きを固定したもとでベクトル場の組 (ξ,η) の取り方に依らない．

最後に退化方向を反転する場合を考える．正に向き付けられた組を保つように $(\xi,\eta)\mapsto(-\xi,-\eta)$ とすればよい．このとき

$$X\mapsto -X, \quad Y\mapsto -Y, \quad Z\mapsto Z, \quad W\mapsto -W$$

である．したがって $\omega_{3,4}$ は変わらず，$\Omega_{3,5}(\xi,\eta)$ と $\Gamma(\xi,\eta)$ はともに符号が反転する．よって $\widehat{\omega}_{3,5}$ も符号が反転する． □



Remark 5.10. 定義域および退化方向の向きを固定する．このとき，$\mathbb{R}^3$ の向きを保つ等長変換のもとで $\widehat{\omega}_{3,5}$ の値は保存され，向きを反転する等長変換のもとでは符号が反転する．

Remark 5.11. $S(f)\cap U$ 上で $\omega_{3,4}\equiv 0$ の場合には，

$$\widehat{\omega}_{3,5}=\omega_{3,5}$$

が成り立つ．一方，$\widehat{\omega}_{3,5}$ は $\omega_{3,4}\equiv 0$ を仮定しない場合にも定義される．したがって，$\widehat{\omega}_{3,5}$ は従来の $(3,5)$-カスプ的曲率を拡張した量である．

5.3 積曲率の導入と特異点型の内在的判定

本 subsection では，ジェネリックな特異点における内在的判定条件を，$\kappa^{\mathrm{int}}_{3,4},\kappa^{\mathrm{int}}_{3,5}$ を用いて記述する．

Theorem 5.12 (ジェネリックな特異点型の内在的判定)**.** $f:\Sigma\to\mathbb{R}^3$ をフロンタル，$p\in\Sigma$ とし，f は p で 3-*type edge* であるとする．さらに，f は p でジェネリックであるとする．このとき，次の条件は同値である．

(1) f は p で $(3,4)$-カスプ辺である．

(2) $\kappa^{\mathrm{int}}_{3,4}(p)\neq 0$ である．

また，次の条件も同値である．

(1) f は p で $(3,5)$-カスプ辺である．

(2) p の近傍 U が存在して，$S(f)\cap U$ 上で $\kappa^{\mathrm{int}}_{3,4}\equiv 0$ であり，かつ $\kappa^{\mathrm{int}}_{3,5}(p)\neq 0$ である．

Example 5.13 (ジェネリック仮定を除けないことを示す例)*.* 次の写像を考える：

$$f(u,v)=\left(u,\ \frac{u^2}{2}+\frac{v^3}{6},\ \frac{v^4}{24}\right).$$

この写像は原点で $(3,4)$-カスプ辺であるが，特異曲線上で $\det(f_u,\ f_{v^3},\ f_{uu})=0$ であるから，補題 3.13 より $\kappa^{\mathrm{int}}_{3,4}\equiv 0$ を得る．また，直接計算より符号付極限法曲率も $\widetilde{\kappa}_\nu\equiv 0$ である．すなわち，非ジェネリックな場合には，$(3,4)$-カスプ辺であっても $\kappa^{\mathrm{int}}_{3,4}=0$ となり得る．したがって，定理 5.12 におけるジェネリック仮定は除くことができない．

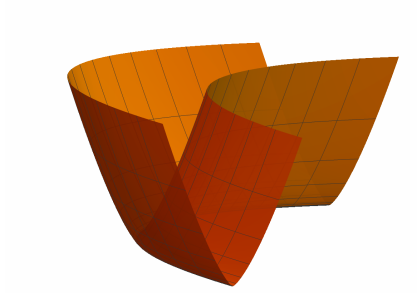


Figure 6: $f(u,v)=\left(u,\frac{u^2}{2}+\frac{v^3}{6},\frac{v^4}{24}\right)$ の像．特異曲線上で $\kappa^{\mathrm{int}}_{3,4}=0$ であるが，特異点はすべて $(3,4)$-カスプ辺である．

定理 5.12 の証明のために，カスプ的曲率 $\omega_{3,4}$ と補正されたカスプ的曲率 $\widehat{\omega}_{3,5}$ に符号付き極限法曲率 $\widetilde{\kappa}_\nu$ を掛けた積曲率 $\kappa^{\Pi}_{3,4},\kappa^{\Pi}_{3,5}$ を導入して，$\kappa^{\mathrm{int}}_{3,4},\kappa^{\mathrm{int}}_{3,5}$ との関係を述べる (命題 5.15, 5.16)．

Definition 5.14 (積曲率)*.* $f:\Sigma\to\mathbb{R}^3$ をフロンタル，$p\in\Sigma$ とし，f は p で 3-type edge であるとする．事実 5.7 および定義 5.2 の $\omega_{3,4}$ と $\widetilde{\kappa}_\nu$ を用いて

$$\kappa^{\Pi}_{3,4}(p):=\widetilde{\kappa}_\nu(p)\omega_{3,4}(p)$$

と定め，これを $(3,4)$-積曲率という．また，定義 5.8 を用いて

$$\kappa^{\Pi}_{3,5}(p):=\widetilde{\kappa}_\nu(p)\widehat{\omega}_{3,5}(p)$$

と定め，これを $(3,5)$-積曲率という．$\omega_{3,4}\equiv 0$ はこの定義のために必要ない．

Proposition 5.15 ($\kappa^{\mathrm{int}}_{3,4}$ と $(3,4)$-積曲率)**.** $f:\Sigma\to\mathbb{R}^3$ をフロンタル，$p\in\Sigma$ とし，f は p で 3-*type edge* であるとする．特異曲線上で

$$\kappa^{\Pi}_{3,4}=\frac{1}{2}\kappa^{\mathrm{int}}_{3,4}$$

が成り立つ．特に，$\kappa^{\Pi}_{3,4}$ は内在的不変量である．

Proof. p の十分小さな開近傍 U において，

$$S(f)\cap U=\{v=0\},\qquad \xi=\partial_u\quad\text{on } S(f)\cap U,\qquad \eta=\partial_v\quad\text{on } U$$

を満たす局所座標 (u,v) を取る．定義 5.14 に $\widetilde{\kappa}_\nu$, $\omega_{3,4}$ の定義式を代入することにより，

$$\kappa^{\Pi}_{3,4}=\frac{\det(f_u,f_{v^3},f_{uu})\det(f_u,f_{v^3},f_{v^4})}{\|f_u\|^{2/3}\|f_u\times f_{v^3}\|^{10/3}}$$

である．一方，補題 3.13 より

$$\kappa^{\mathrm{int}}_{3,4}=2\frac{\det(f_u,f_{v^3},f_{uu})\det(f_u,f_{v^3},f_{v^4})}{\|f_u\|^{2/3}\|f_u\times f_{v^3}\|^{10/3}}$$

であることから，主張が従う． □

次に，$(3,5)$-積曲率 $\kappa^{\Pi}_{3,5}$ と $\kappa^{\mathrm{int}}_{3,5}$ の関係を述べる．

Proposition 5.16 ($\kappa^{\mathrm{int}}_{3,5}$ と $(3,5)$-積曲率)**.** $f:\Sigma\to\mathbb{R}^3$ をフロンタル，$p\in\Sigma$ とし，f は p で 3-*type edge* であるとする．p の近傍で *NACS* (x,y) を取り，(∂_x,∂_y) が正に向き付けられるように定義域の向きを取り，$\xi=\partial_x$，$\eta=\partial_y$ とする．単位法ベクトル場 ν の向きを特異曲線上で $\nu=f_x\times f_{y^3}$ となるように選ぶ．$\hat{L}:=\langle f_{xx},\nu\rangle$ とおくと，特異曲線上で

$$\kappa^{\Pi}_{3,5}=\frac{1}{2}\kappa^{\mathrm{int}}_{3,5}-2\hat{L}_y\omega_{3,4}$$

が成り立つ．さらに，$\kappa^{\mathrm{int}}_{3,4}(p)=0$ かつ p がジェネリックならば，

$$\kappa^{\Pi}_{3,5}(p)=\frac{1}{2}\kappa^{\mathrm{int}}_{3,5}(p)$$

である．

Proof. 因子の補題により，ある滑らかな写像 φ を用いて $f_y=y^2\varphi$ と書ける．関数 $\mathcal{G}$, $\mathcal{M}$, $\mathcal{N}$ をそれぞれ

$$\mathcal{G}:=\langle\varphi,\varphi\rangle,\qquad \mathcal{M}:=\langle\varphi_x,\nu\rangle,\qquad \mathcal{N}:=\langle\varphi_y,\nu\rangle$$

と置くと，$\langle\varphi,\nu\rangle=0$ より

$$\hat{G}=y^4\mathcal{G},\qquad \hat{M}=y^2\mathcal{M},\qquad \hat{N}=y^2\mathcal{N}$$

が成り立つ．ただし，

$$\hat{L}=\langle f_{xx},\nu\rangle,\quad \hat{M}:=\langle f_{xy},\nu\rangle,\quad \hat{N}:=\langle f_{yy},\nu\rangle$$

とする．また，NACS では $\hat{F}\equiv0$ であるから，正則点上で

$$\widetilde{K}=y^2K=\frac{\hat{L}\mathcal{N}-y^2\mathcal{M}^2}{\hat{E}\mathcal{G}}\tag{5.5}$$

である．

以降，特に断らない限り，すべての量を $(x,0)$ で評価する．特異曲線上では $\hat{E}=1$, $\hat{E}_y=0$, $\mathcal{G}=1/4$ である．したがって，式 (5.5) を y で微分して $y=0$ とすると，

$$\widetilde{K}_y=4\big(\hat{L}_y\mathcal{N}+\hat{L}\mathcal{N}_y\big)-16\hat{L}\mathcal{N}\mathcal{G}_y\tag{5.6}$$

が成り立つ．また，$f_y=y^2\varphi$ より，

$$f_{y^3}=2\varphi,\qquad f_{y^4}=6\varphi_y,\qquad f_{y^5}=12\varphi_{yy},\qquad \widetilde{\kappa}_\nu=\hat{L}$$

である．補正された $(3,5)$-カスプ的曲率の定義と NACS の定義から，

$$\widehat{\omega}_{3,5} = \det(f_x, f_{y^3}, f_{y^5}) - \frac{5}{3}\Gamma\omega_{3,4} = 12\langle\varphi_{yy}, \nu\rangle - \frac{5}{3}\Gamma\omega_{3,4}$$

を得る．よって，

$$\kappa^{\Pi}_{3,5} = 12\hat{L}\langle\varphi_{yy}, \nu\rangle - \frac{5}{3}\Gamma\kappa^{\Pi}_{3,4} \tag{5.7}$$

が成り立つ．一方，特異曲線上では $f_{xy} = 0$ であるから，$\langle\nu_y, f_x\rangle = 0$ が成り立つ．また，ν が単位法ベクトル場であることから，$\langle\nu_y, \nu\rangle = 0$ である．さらに，$\hat{F} \equiv 0$ および $f_y = y^2\varphi$ より，$\langle f_x, \varphi\rangle = \langle\nu, \varphi\rangle = 0$ が成り立つ．したがって，ν_y は φ に平行である．さらに，

$$\langle\nu_y, \varphi\rangle = -\langle\nu, \varphi_y\rangle = -\mathcal{N}, \qquad \|\varphi\|^2 = \mathcal{G} = \frac{1}{4}$$

より，$\nu_y = -4\mathcal{N}\varphi$ となる．よって $\langle\varphi_y, \nu_y\rangle = -4\mathcal{N}\langle\varphi_y, \varphi\rangle = -2\mathcal{G}_y\mathcal{N}$ であり，

$$\mathcal{N}_y = \langle\varphi_{yy}, \nu\rangle + \langle\varphi_y, \nu_y\rangle = \langle\varphi_{yy}, \nu\rangle - 2\mathcal{G}_y\mathcal{N} \tag{5.8}$$

が成り立つ．式 (5.8) を式 (5.6) へ代入し，さらに $\widetilde{K} = 4\hat{L}\mathcal{N}$, $\omega_{3,4} = 6\mathcal{N}$ および式 (5.7) を用いて整理すると，

$$\kappa^{\Pi}_{3,5} = 3\widetilde{K}_y - 2\hat{L}_y\omega_{3,4} + 18\widetilde{K}\,\mathcal{G}_y - \frac{5}{3}\Gamma\kappa^{\Pi}_{3,4} \tag{5.9}$$

を得る．

ここで，NACS の定義と $\hat{F} \equiv 0$ より，

$$\Gamma = \langle f_{y^3}, f_{y^4}\rangle = 12\langle\varphi, \varphi_y\rangle = 6\mathcal{G}_y$$

が成り立つ．また，

$$\kappa^{\Pi}_{3,4} = \hat{L}\omega_{3,4} = 6\hat{L}\mathcal{N} = \frac{3}{2}\widetilde{K}$$

である．したがって，$5\Gamma\kappa^{\Pi}_{3,4}/3 = 15\widetilde{K}\,\mathcal{G}_y$ となるため，式 (5.9) に代入して

$$\kappa^{\Pi}_{3,5} = 3\widetilde{K}_y - 2\hat{L}_y\omega_{3,4} + 3\widetilde{K}\,\mathcal{G}_y \tag{5.10}$$

を得る．補題 3.16 より

$$\widetilde{K}(x,0) = \frac{1}{3}\kappa^{\mathrm{int}}_{3,4}(x), \qquad \widetilde{K}_y(x,0) = \frac{1}{6}\kappa^{\mathrm{int}}_{3,5}(x) - \frac{\hat{G}_{y^5}(x,0)}{360}\kappa^{\mathrm{int}}_{3,4}(x)$$

であり，また，$\hat{G} = y^4\mathcal{G}$ を 5 階微分して $y = 0$ とすると $\hat{G}_{y^5} = 120\mathcal{G}_y$ が成り立つため，これらを式 (5.10) へ代入すると，

$$\kappa^{\Pi}_{3,5} = \frac{1}{2}\kappa^{\mathrm{int}}_{3,5} - \frac{\hat{G}_{y^5}}{120}\kappa^{\mathrm{int}}_{3,4} - 2\hat{L}_y\omega_{3,4} + \frac{\hat{G}_{y^5}}{120}\kappa^{\mathrm{int}}_{3,4} = \frac{1}{2}\kappa^{\mathrm{int}}_{3,5} - 2\hat{L}_y\omega_{3,4}$$

を得る．さらに，$\kappa^{\mathrm{int}}_{3,4}(p) = 0$ と p におけるジェネリック性から $\widetilde{\kappa}_\nu(p) \neq 0$ である．命題 5.15 より

$$0 = \frac{1}{2}\kappa^{\mathrm{int}}_{3,4}(p) = \kappa^{\Pi}_{3,4}(p) = \widetilde{\kappa}_\nu(p)\omega_{3,4}(p)$$

なので，$\omega_{3,4}(p) = 0$ を得る．ゆえに $\kappa^{\Pi}_{3,5}(p) = \kappa^{\mathrm{int}}_{3,5}(p)/2$ となり，主張が従う． □

Remark 5.17. 従来の $(3,5)$-カスプ的曲率 $\omega_{3,5}$ は，低次消滅条件 $\omega_{3,4} \equiv 0$ のもとで well-defined となる．このとき，$\kappa^{\mathrm{int}}_{3,5} = 2\widetilde{\kappa}_\nu\omega_{3,5}$ が成り立つ．これに対して，$\kappa^{\mathrm{int}}_{3,5}$ はこの条件を課すことなく定義することができる．したがって，$\kappa^{\mathrm{int}}_{3,5}$ は従来の $\omega_{3,5}$ に対応する内在的な情報を，より一般の 3-type edge に対して与える量とみなすことができる．

定理 *5.12* の証明. 命題 5.15 より

$$\kappa^{\mathrm{int}}_{3,4}(p) \neq 0 \quad \Longleftrightarrow \quad \kappa^{\Pi}_{3,4}(p) \neq 0$$

である．さらに，p におけるジェネリック性 (命題 5.6) より，$\kappa^{\mathrm{int}}_{3,4}(p) \neq 0$ は $\omega_{3,4}(p) \neq 0$ と同値である．[4, Proposition 2.11] より 3-type edge がフロントであることと $\omega_{3,4}(p) \neq 0$ は同値である．定理 4.6 を用いることで，$(3,4)$-カスプ辺に関する同値条件を得る．

次に，$(3,5)$-カスプ辺に関する同値条件を示す．p におけるジェネリック性と命題 5.15 より，p の十分小さな近傍では $\kappa^{\mathrm{int}}_{3,4} \equiv 0$ と $\omega_{3,4} \equiv 0$ は同値である．さらに，[4, Proposition 2.11] より，これは特異曲線上の近傍全体で非フロントであることと同値である．さらに命題 5.16 と注意 5.11 より，$\omega_{3,4} \equiv 0$ のもとでは

$$\kappa^{\mathrm{int}}_{3,5}(p) \neq 0 \iff \kappa^{\Pi}_{3,5}(p) \neq 0 \iff \widehat{\omega}_{3,5}(p) \neq 0 \iff \omega_{3,5}(p) \neq 0$$

が成り立つ．さらに，$\omega_{3,5}$ の定義より，$\omega_{3,5}(p) \neq 0$ は $\det(\xi f, \eta^3 f, \eta^5 f)(p) \neq 0$ と同値である．定理 4.11 から，$(3,5)$-カスプ辺に関する同値条件が従う． □

命題 3.12 では，$(3,4)$-カスプ辺の形状について $\kappa^{\mathrm{int}}_{3,4}$ の符号による十分条件を与えた．一方，定理 5.12 より，ジェネリックな $(3,4)$-カスプ辺では $\kappa^{\mathrm{int}}_{3,4}(p) \neq 0$ が成り立つ．したがって，対象の点がジェネリックな場合にはこれらの十分条件が必要十分条件となり，次の系を得る．

こっちを主結果にしてよい(?)

Corollary 5.18. $f : \Sigma \to \mathbb{R}^3$ をフロンタル，$p \in \Sigma$ とし，f は p で $(3,4)$-カスプ辺であるとする．さらに，f は p でジェネリックであるとする．このとき，次が成り立つ．

(1) $\kappa^{\mathrm{int}}_{3,4}(p) > 0$ であることと，f が p で凸であることは同値である．

(2) $\kappa^{\mathrm{int}}_{3,4}(p) < 0$ であることと，f が p で鞍状であることは同値である．

Proof. 定理 5.12 より，$\kappa^{\mathrm{int}}_{3,4}(p) \neq 0$ が成り立つ．一方，命題 3.12 より，$\kappa^{\mathrm{int}}_{3,4}(p) > 0$ ならば f は p で凸であり，$\kappa^{\mathrm{int}}_{3,4}(p) < 0$ ならば f は p で鞍状である．凸と鞍状は両立しないため，$\kappa^{\mathrm{int}}_{3,4}(p) \neq 0$ と合わせて逆の含意も成り立つ．よって，主張が従う． □

5.4 系 D の証明

定理 5.12 と第 4.1 節の結果を用いて，ジェネリックな $(3,4)$-カスプ辺と $(3,5)$-カスプ辺は property-$\mathcal{A}$ (式 (1.1)) を満たすことを示す．

Corollary 5.19 (等長性による特異点型の保存)**.** $f : \Sigma \to \mathbb{R}^3$，$g : \overline{\Sigma} \to \mathbb{R}^3$ を等長的なフロンタルとし，$p \in \Sigma$，$\overline{p} \in \overline{\Sigma}$ を対応するジェネリックな特異点とする．このとき，$j = 4, 5$ に対して，次の条件は同値である．

(1) f は p で $(3,j)$-カスプ辺である．

(2) g は $\overline{p}$ で $(3,j)$-カスプ辺である．

Proof. 第一種性と退化次数 2 特異点は内在的な性質であるから，命題 4.4 より，3-type edge であることは等長性のもとで保存される．また，$\kappa^{\mathrm{int}}_{3,4}, \kappa^{\mathrm{int}}_{3,5}$ は内在的不変量であるため，$\kappa^{\mathrm{int}}_{3,4}(p) \neq 0$，$\kappa^{\mathrm{int}}_{3,5}(p) \neq 0$，および特異曲線の十分小さい近傍で $\kappa^{\mathrm{int}}_{3,4} \equiv 0$ となる条件も等長性のもとで保存される．よって，両点におけるジェネリック性と定理 5.12 より，$j = 4, 5$ に対して主張が従う． □

Appendix.

A 命題 4.3 の証明

定義域の座標変換に対しては，符号付き面積密度関数は非零の滑らかな関数倍だけ変化することは明らかである．したがって，次の補題 A.1 を示せばよい．

Lemma A.1 (値域微分同相による面積密度の変換)**.** $U \subset \Sigma$ を開集合，$f : U \to \mathbb{R}^3$ をフロンタル，$\nu : U \to \mathbb{S}^2$ を単位法ベクトル場とする．$\Phi : (\mathbb{R}^3, f(p)) \to (\mathbb{R}^3, \Phi(f(p)))$ を局所微分同相写像芽，$\widetilde{f} := \Phi \circ f$ とする．各 $q \in U$ に対して $J(q) := d\Phi_{f(q)}$ と置き，

$$\widetilde{\nu}(q) := \frac{(J(q)^T)^{-1}\nu(q)}{\|(J(q)^T)^{-1}\nu(q)\|}$$

と定める．このとき，f と $\widetilde{f}$ の符号付き面積密度関数 $\lambda, \widetilde{\lambda}$ は

$$\widetilde{\lambda} = \det J \, \|(J^T)^{-1}\nu\| \, \lambda$$

を満たす．したがって，値域の局所微分同相は特異点集合と退化次数を保つ．

Proof. $f_u \times f_v = \lambda\nu$ であり，J に対して

$$(Ja) \times (Jb) = \det J\,(J^T)^{-1}(a \times b)$$

が成り立つ．よって

$$\widetilde{\lambda} = \left\langle (Jf_u) \times (Jf_v), \frac{(J^T)^{-1}\nu}{\|(J^T)^{-1}\nu\|} \right\rangle = \det J\,\|(J^T)^{-1}\nu\|\,\lambda.$$

右辺の λ の係数は非零の滑らかな関数であるから，零点集合と零の次数は保たれる． □

B　定理 4.11 の証明

定理 *4.11* の証明．まず (1) ⇒ (2) を示す．標準的な $(3,5)$-カスプ辺 $f_0(u,v) := (v^3, v^5, u)$ を考える．

$$(f_0)_u \times (f_0)_v = v^2(-5v^2, 3, 0)$$

であるから，$S(f_0) = \{v = 0\}$ である．また，f_0 は 3-type edge の標準形と $\mathcal{A}$-同値であり，特異点集合上では退化方向が ∂_v，特異方向が ∂_u である．$\xi_0 := \partial_u$, $\eta_0 := \partial_v$ と取ると，$v = 0$ 上で

$$\xi_0 f_0 = (0,0,1), \quad \eta_0^3 f_0 = (6,0,0), \quad \eta_0^4 f_0 = \mathbf{0}, \quad \eta_0^5 f_0 = (0,120,0).$$

したがって，直接計算より

$$\det(\xi_0 f_0, \eta_0^3 f_0, \eta_0^4 f_0) \equiv 0, \qquad \det(\xi_0 f_0, \eta_0^3 f_0, \eta_0^5 f_0) \neq 0$$

である．[4, Proposition 2.11] より，f_0 は特異曲線上の各点でフロントではない．よって，f_0 は条件 (2) を満たす．次に，一般の $(3,5)$-カスプ辺 f でも (2) が成り立つことを示す．フロントでないことは $\mathcal{A}$-同値のもとで保存される．また，3-type edge に対して，特異曲線上で $\det(\xi f, \eta^3 f, \eta^4 f) \equiv 0$ であること，およびそのもとで $\det(\xi f, \eta^3 f, \eta^5 f)(p) \neq 0$ であることは，ベクトル場の組 (ξ, η) の取り方によらず，$\mathcal{A}$-同値のもとで保存される（[4, Lemmas 2.3, 2.6]）．したがって，任意の $(3,5)$-カスプ辺は条件 (2) を満たす．

次に (2) ⇒ (1) を示す．補題 4.8 より，定義域と値域の局所座標を適当に取れば

$$f(u,v) = \big(v^3,\, v^4 b_1(u) + v^5 b_2(u) + v^6 b_3(u,v),\, u\big) \tag{B.1}$$

と書ける．ここで，非フロント性のもとでは，条件 (2) の行列式の非零性は，特異ベクトル場と退化ベクトル場の取り方，および定義域と値域の局所座標変換のもとで保存される（[4, Lemmas 2.3, 2.6]）．特異曲線上で $\xi = \partial_u$，$\eta = \partial_v$ と取る．$f_v/v^2 \times f_u$ から得られる単位法ベクトル場 ν は，$v = 0$ 上で

$$\nu_v(u,0) = \left(\frac{4}{3} b_1(u), 0, 0\right)$$

を満たす．$f_v(u,0) = \mathbf{0}$ であるから，特異曲線上の非フロント性より $b_1 \equiv 0$ を得る．また，直接計算により

$$\det(f_u, f_{v^3}, f_{v^5})(0,0) = 720 b_2(0)$$

であるから，仮定より $b_2(0) \neq 0$ を得る．

以下では，これらの条件のもとで f を標準的な $(3,5)$-カスプ辺の形へ変換していく．まず，値域の座標変換 $(X,Y,Z) \longmapsto (X, Y/b_2(Z), Z)$ を施し，b_3/b_2 を $v = 0$ のまわりで展開すると，式 (B.1) は

$$f(u,v) = \big(v^3, v^5 + v^6 c_1(u) + v^7 c_2(u) + v^8 c_3(u,v), u\big) \tag{B.2}$$

と書ける．ここで定義域の座標変換として

$$v = \widetilde{v} - \frac{c_2(u)}{5}\widetilde{v}^3 - \frac{c_2(u)^2}{25}\widetilde{v}^5$$

を用いる．Taylor 展開すると，滑らかな関数 A, B を用いて式 (B.2) の f の第 1 成分 f_1 と第 2 成分 f_2 は

$$\begin{aligned} f_1(u,\widetilde{v}) &= \widetilde{v}^3 - \frac{3c_2(u)}{5}\widetilde{v}^5 + \widetilde{v}^8 A(u,\widetilde{v}), \\ f_2(u,\widetilde{v}) &= \widetilde{v}^5 + c_1(u)\widetilde{v}^6 + \widetilde{v}^8 B(u,\widetilde{v}). \end{aligned} \tag{B.3}$$

と表せる．$\widetilde{v}$ を再び v と書き，A, B に対してホイットニーの補題を 3 回適用する．滑らかな関数 A_j, B_j（$j = 0, \dots, 7$）を用いて

$$A(u,v) = \sum_{j=0}^{7} v^j A_j(v^8, u), \qquad B(u,v) = \sum_{j=0}^{7} v^j B_j(v^8, u)$$

と書ける．$X = v^3$, $Y = v^5$, $Z = u$ と置くと，

$$v^8 = XY, \qquad v^9 = X^3, \qquad v^{10} = Y^2, \qquad v^{11} = X^2Y,$$
$$v^{12} = X^4, \qquad v^{13} = XY^2, \qquad v^{14} = X^3Y, \qquad v^{15} = Y^3$$

である．そこで値域上の滑らかな関数 P, Q を

$$\begin{aligned} P(X,Y,Z) &:= XYA_0(XY,Z) + X^3A_1(XY,Z) + Y^2A_2(XY,Z) + X^2YA_3(XY,Z) \\ &\quad + X^4A_4(XY,Z) + XY^2A_5(XY,Z) + X^3YA_6(XY,Z) + Y^3A_7(XY,Z), \\ Q(X,Y,Z) &:= XYB_0(XY,Z) + X^3B_1(XY,Z) + Y^2B_2(XY,Z) + X^2YB_3(XY,Z) \\ &\quad + X^4B_4(XY,Z) + XY^2B_5(XY,Z) + X^3YB_6(XY,Z) + Y^3B_7(XY,Z) \end{aligned}$$

と定める．このとき

$$P(v^3, v^5, u) = v^8A(u,v), \qquad Q(v^3, v^5, u) = v^8B(u,v)$$

が成り立つ．したがって，値域の座標変換を

$$\Phi(X,Y,Z) := \left(X - \frac{3c_2(Z)}{5}Y + P(X,Y,Z),\ Y + X^2c_1(Z) + Q(X,Y,Z),\ Z\right)$$

と置けば，式 (B.3) より標準形 $f_0(u,v) = (v^3, v^5, u)$ に対して

$$\begin{aligned} \Phi \circ f_0(u,v) &= \left(v^3 - \frac{3c_2(u)}{5}v^5 + v^8A(u,v),\ v^5 + c_1(u)v^6 + v^8B(u,v),\ u\right) \\ &= f(u,v) \end{aligned}$$

となる．また，$(dP)_{(0,0,0)} = (dQ)_{(0,0,0)} = 0$ であることから

$$d\Phi_{(0,0,0)} = \begin{pmatrix} 1 & -3c_2(0)/5 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det(d\Phi_{(0,0,0)}) = 1 \neq 0$$

であり，逆関数定理より Φ は原点近傍の局所微分同相である．よって元の写像芽は f_0 と $\mathcal{A}$-同値であり，p は $(3,5)$-カスプ辺である． □

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Data availability

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

Conflict of interest

The authors declare that they have no conflict of interest.

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